CHARUSAT
CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
Accredited Grade A by NAAC

ACADEMIC REGULATIONS & SYLLABUS

Faculty of Sciences
P. D. Patel Institute of Applied Sciences
Department of Chemical Sciences
Master of Science Programme
(Advanced Organic Chemistry)

P. D. Patel Institute of APPLIED SCIENCES

CHAROTARUNIVERSITY OF SCIENCE AND TECHNOLOGY
Faculty of Applied Sciences

ACADEMIC REGULATIONS

M. Sc. (Advanced Organic Chemistry) Programme

Charotar University of Science and Technology (CHARUSAT)
CHARUSAT Campus, At Post: Changa – 388421, Taluka: Petlad, District: Anand
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Year – 2022

CHARUSAT

FACULTY OF APPLIED SCIENCES
ACADEMIC REGULATIONS
M. Sc. (Advanced Organic Chemistry) Programme

To ensure uniform system of education, duration of post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following are the academic rules and regulations.

1. *System of Education*

The Semester system of education shall be followed across The Charotar University of Science and Technology (CHARUSAT) at Master's levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to do a specified course work in the chosen subject of specialization and also complete a project/dissertation if any. Medium of instruction will be English

2. *Duration of Programme*

Postgraduate programme	(M.Sc.)
Minimum	4 semesters (2 academic years)
Maximum	6 semesters (3 academic years)

The maximum limit can be extended by 1 or 2 semester subject to the approval of university on case to case basis.

3. *Eligibility for admissions*

For the admission to M.Sc., programs in the subject of Biological/Physical/Mathematical/Chemical Sciences a candidate must have obtained a Degree of Bachelor of Science from any recognized University or a Degree recognized as equivalent thereto, with minimum Second Class.

4. *Mode of admissions*

Admission to M.Sc. programme will purely on combined merit of admission test and performance at graduation.

5. Programme structure and Credits

A student admitted to a program should study the course and earn credits specified in the course structure. (Please refer Annexure-A)

6. Attendance

6.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

6.2 Student attendance in a course should be 80%.

7. Course Evaluation

7.1 The performance of every student in each course will be evaluated as follows:

7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and

7.1.2 Final examination will be conducted by the University t for 70% of the marks for the course.

7.2 *Internal Evaluation*

7.2.1 Internal evaluation will be based on internal tests and several other tools of assessment like, quiz, viva, seminar etc., as prescribed by concerned teacher and decided by the faculty.

7.3 Internal Institutional evaluation for practical

7.3.1 One internal practical test/viva will be conducted per semester totaling to 30 % internal marks for practical

7.3.2 In “Continuous evaluation” Students shall be evaluated in a continuous manner for their involvement in the practical, aptitude for learning, completion of practical related assignments, regularity in the practical and record keeping

7.4 University Examination

7.4.1 The final examination by the University for 70% of the evaluation for the course will be through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these.

7.4.2 In order to earn the credit in a course a student has to obtain grade other than FF.

7.5 Performance at Internal & University Examination

7.5.1 Minimum performance with respect to internal marks as well as university examination will be an important consideration for passing a course.
Details of minimum percentage of marks to be obtained in the examinations are as follows

Minimum marks in University Exam per subject	Minimum marks Overall per subject
40%	50%

7.5.1. If a candidate obtains minimum required marks per subject but fails to obtain minimum required overall marks, he/she has to repeat the university examination till the minimum required overall marks are obtained.(As per the clause 8.2(iv)

8 Grading

8.1 The internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

Grading Scheme:

Range of Marks (%)	≥80	≥75 <80	≥70 <75	≥65 <70	≥60 <65	≥55 <60	≥50 <55	<50
Letter Grade	AA	AB	BB	BC	CC	CD	DD	FF
Grade Point	10	9	8	7	6	5	4	0

8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

(i) SGPA = $\sum C_i G_i / \sum C_i$ where C_i is the number of credits of course i
G_i is the Grade Point for the course i
and i = 1 to n, n = number of courses in the semester

(ii) CGPA = $\sum C_i G_i / \sum C_i$ where C_i is the number of credits of course i
G_i is the Grade Point for the course i
and i = 1 to n, n = number of courses of all semesters up to which CGPA is computed.

(iii) No student will be allowed to move further if CGPA is less than 3 at the end of every academic year.

9. Awards of Degree

9.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

- 9.1.1 He should have earned at least minimum required credits as prescribed in course structure; and
- 9.1.2 He should have cleared all internal and external evaluation components in every course; and
- 9.1.3 He should have secured a minimum CGPA of 5.0 at the end of the programme;
- 9.1.4 In addition to above, the student has to complete the required formalities as per the regulatory bodies.

9.2 The student who fails to satisfy minimum requirement of CGPA will be allowed to improve the grades so as to secure a minimum CGPA for award of degree. Only latest grade will be considered.

10 Award of Class:

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Award of Class	**CGPA Range**
First Class with Distinction	CGPA ≥ 7.50 & ≤ 10.00
First class	CGPA ≥ 6.00 & <7.50
Second Class	CGPA ≥ 5.00 & <6.00
Pass Class	CGPA<5.00

11 Transcript:

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA,CGPA, class obtained, etc.

VISION

To become an eminent national institute imparting science education integrated with research.

MISSION

To engage in education, research, and spread of science for the benefit of society.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

The Postgraduate would be able to

PEO-1: Apply the acquired knowledge in various fields of sciences to plan and execute tasks. (POs 1, 2, 3, 8)

PEO-2: Have scientific aptitude and competency in solving local and global problems. (POs 4, 6, 9, 10)

PEO-3 Possess an ethical approach, while executing research projects and scientific writing. (POs 7, 9, 10)

PEO-4 Ability to follow life-long learning. (POs 11)

PEO-5 Understand the importance of team work and possess leadership and entrepreneurship traits. (POs 5, 6, 8)

N.B. POs are mapped with PEOs.

PROGRAM OUTCOMES (POs)

The Programme Outcome are as under:

The Post Graduates would be able to

1. **Knowledge in sciences:** Possess knowledge and comprehend the various core and allied courses offered under the opted stream of sciences.
2. **Planning Abilities:** Demonstrate effective planning abilities, including time management, resource management, delegation skills, organizational skills, teamwork, and interpersonal skills.
3. **Problem analysis:** Utilize the principles of scientific enquiry and think analytically, critically with clarity of thought, while solving problems.

4. **Modern tool usage:** **A**pply appropriate methodologies, methods and procedures, resources to conceptualize the matters.
5. **Leadership skills:** Realize the importance of teamwork and take initiatives to plan and implement a workflow to fulfill tasks with a compassionate attitude.
6. **Professional Identity:** Establish as a professional in the chosen area of the profession with social responsibility**.**
7. **Ethics in Science:** To understand the value of being ethical and implement its principles while carrying out research and publication, tasks at workplace and in life. To respect and adhere to the rules/regulations/treaties/directives/notifications/laws related to scientific research and business.
8. **Communication:** Communicate effectively and responsibly with the scientific community and with society at large, such as being able to comprehend and write effective reports, make effective presentations and documentation, and give and receive clear instructions.
9. **Modern Science and Society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety and legal issues and help society whenever situations arise.
10. **Environment and sustainability:** Understand the causes of environmental pollution and how it can be reduced and remedied using green technology in industries and environment.
11. **Life-long learning:** Vigilant about changes in the worldwide professional scenario arising out of technological, political, and economic factors. Recognize the need for and undertake steps necessary to fill the gaps recognized to keep oneself professionally competent. Self- assess and use feedback effectively from others to identify learning needs and to satisfy these needs on an ongoing basis.

CHOICE BASED CREDIT SYSTEM

With the aim of incorporating the various guidelines initiated by the University Grants Commission (UGC) to bring equality, efficiency and excellence in the Higher Education System, Choice Based Credit System (CBCS) has been adopted. CBCS offers wide range of choices to students in all semesters to choose the courses based on their aptitude and career objectives. It accelerates the teaching-learning process and provides flexibility to students to opt for the courses of their choice and / or undergo additional courses to strengthen their Knowledge, Skills and Attitude.

1. CBCS – Conceptual Definitions / Key Terms (Terminologies)

Types of Courses: The Programme Structure consist of 4 types of courses: Foundation courses, Core courses, Elective courses and Non-credit (audit) courses.

1.1. Foundation Course

These courses are offered by the institute in order to prepare students for studying courses to be offered at higher levels.

1.2. Core Courses

A Course which shall compulsorily be studied by a candidate to complete the requirements of a degree / diploma in a said programme of study is defined as a core course. Following core courses are incorporated in CBCS structure:

A. University Core courses(UC):

University core courses are compulsory courses which are offered across university and must be completed in order to meet the requirements of programme. Environmental science will be a compulsory University core for all Undergraduate Programmes.

B. Programme Core courses(PC):

Programme core courses are compulsory courses offered by respective programme owners, which must be completed in order to meet the requirements of programme.

1.3. Elective Courses

Generally, a course which can be chosen from a pool of courses and which may be very specific or specialised or advanced or supportive to the discipline of study or which provides an extended scope or which enables an exposure to some other discipline / domain or nurtures the candidates proficiency / skill is called an elective course. Following elective courses are incorporated in CBCS structure:

A. University Elective Courses(UE):

The pool of elective courses offered across all faculties / programmes. As a general guideline, Programme should incorporate 2 University Electives of 2 credits each (total 4 credits).

B. Institute Elective Course (IE)

Institute ekective courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialisation

C. Programme Elective Courses(PE):

The programme specific pool of elective courses offered by respective programme.

D. Cluster Elective Course (CE):

An ‘Elective Course’ is a course which students can choose from the given set of functional course/ Area or Streams of Specialization options (eg. Common Courses to EC/CE/IT/EE) as offered or decided by the Institute from time-to-time.

1.4. Non Credit Course (NC) - AUDIT Course

A ‘Non Credit Course’ is a course where students will receive Participation or Course Completion certificate. This will be reflected in Student’s Grade Sheet but the grade of the course will not be consider to calculate SGPA and CGPA. Attendance and Course Assessment is compulsory for Non Credit Courses.

1.5. Medium of Instruction

The Medium of Instruction will be English.

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
P. D. PATEL INSTITUTE OF APPLIED SCIENCES
DEPARTMENT OF CHEMICAL SCIENCES

Degree: M.Sc. (AOC)

Semester I

Course code	Course Title	Teaching Scheme				Examination Scheme				
		Contact hours/week			Credits	TH		PR		Total
		L	T	P		Internal	External	Internal	External	
OC761	Selected topics of Quantum Chemistry	3	1		4	0/30	28/70			100
OC762	Stereochemistry and Organic Reactions	3	1		4	0/30	28/70			100
OC763	Chemical Dynamics, Electrochemistry and Group Theory	3	1		4	0/30	28/70			100
OC764	Separation Techniques	3	1		4	0/30	28/70			100
OC765	Organic Chemistry Laboratory-I			4	2			0/30	28/70	100
OC766	Inorganic Chemistry Laboratory-I			4	2			0/30	28/70	100
OC767	Physical Chemistry Laboratory-I			4	2			0/30	28/70	100
HS*				2	2			0/30	28/70	100

*Course chosen by student from the list of courses offered by Humanities and Social Sciences Department, Faculty of Management studies
**Course chosen by student from the list of University Elective courses

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
P. D. PATEL INSTITUTE OF APPLIED SCIENCES
DEPARTMENT OF CHEMICAL SCIENCES

Degree: M.Sc. (AOC)
Semester II

Course code	Course Title	Teaching Scheme				Examination Scheme				
		Contact hours/week			Credits	TH		PR		Total
		L	T	P		Internal	External	Internal	External	
OC771	Spectroscopy-I	3	1		4	0/30	28/70			100
OC772	Name Reactions and Regents in Organic Chemistry	3	1		4	0/30	28/70			100
OC773	Topics in Physical Chemistry	3	1		4	0/30	28/70			100
OC774	Coordination Chemistry and Magneto Chemistry	3	1		4	0/30	28/70			100
OC775	Organic Chemistry Laboratory-II			4	2			0/30	28/70	100
OC776	Inorganic Chemistry Laboratory-II			4	2			0/30	28/70	100
OC777	Physical Chemistry Laboratory-II			4	2			0/30	28/70	100
HS106.02 (E):	ACADEMIC WRITING			2	2			0/30	28/70	100
UE**				2	2			0/30	28/70	100

*Course chosen by student from the list of courses offered by Humanities and Social Sciences Department, Faculty of Management studies
**Course chosen by student from the list of University Elective courses

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
P. D. PATEL INSTITUTE OF APPLIED SCIENCES
DEPARTMENT OF CHEMICAL SCIENCES

Degree: M.Sc. (AOC)

Semester III

Subject Code	Subjects	Teaching scheme				Theory Evaluation			Practical Evaluation			Total I +II
		L+T	P	Contacts hrs.	Total Credits	Institute	University	Total-I	Institute	University	Total-II	
OC 841	Applications of NMR and ESR Spectroscopy	4		4	4	30	70	100				100
OC 842	Natural Product Chemistry	4		4	4	30	70	100				100
OC 843	Concepts of Heterocyclic Chemistry	4		4	4	30	70	100				100
OC 844	Disconnection approach for organic synthesis	4		4	4	30	70	100				100
OC 845	Practical		14	14	7				90	210	300	300
OC 846*	Instrumental Methods of Analysis-I	2		2	2	15	35	50				50
OC 847*	Advance Problem solving in Chemical Sciences-I	2		2	2	15	35	50				50
					25							750

* Students will opt any one course from OC 846 & OC 847

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
P. D. PATEL INSTITUTE OF APPLIED SCIENCES
DEPARTMENT OF CHEMICAL SCIENCES

Degree: M.Sc. (AOC)

Semester IV

Subject Code	Subjects	Teaching scheme				Theory Evaluation			Practical Evaluation			Total I + II
		L +T	P	Contacts hrs.	Total Credits	Institute	University	Total-I	Institute	University	Total-II	
OC 851	Medicinal Chemistry	4		4	4	30	70	100				100
OC 852	Chemistry of Synthetic Dyes	4		4	4	30	70	100				100
OC 853	Energy & Wave Function of Molecular Orbitals in Organic Chemistry	4		4	4	30	70	100				100
OC 854	Selected Topics in Organic Chemistry	4		4	4	30	70	100				100
OC 855	Practical		14	14	7				90	210	300	300
OC856 *	Instrumental Methods of Analysis -II	2		2	2	15	35	50				50
OC857 *	Advance Problem solving in Chemical Sciences-II	2		2	2	15	35	50				50
					25							750

* Students will opt any one course from OC 856 & OC 857

Course Code: OC761	**Course Name: Selected topics of Quantum Chemistry**		
Credit: 4	**Semester: `1**		
Contact hours/week	**L**	**T**	**P**
	3	**1**	**-**

A. Outline of the Course

Sr. No.	Title of Unit	Hours
1.	**Fundamentals of Quantum Mechanics**	**13**
2.	**Particle in a 1D and 3D Box and Tunneling**	**11**
3.	**Rotational and vibrational motion of a particle**	**13**
4.	**Hydrogen atom**	**12**
5.	**Many electron atoms & angular momenta**	**11**

B. Detailed Syllabus

Unit 1 Fundamental of Quantum Mechanics
Introduction to quantum chemistry, Postulated of Quantum chemistry, Wave function & Schrödinger wave equation, Heisenberg uncertainty principle, Spin and exclusion principle, expectation value, Operator concepts in QM, Hamiltonian operator; angular momentum operator, Linear operators commutation of operators, Commutative property; momentum operator Angular momentum operators and their commutation relations, shift operators and their commutation relations
Unit 2 Particle in a 1D and 3D Box and Tunnelling
Particle in one-dimensional box, three-dimensional box, Numerical of the particle in 3D box, Normalization and orthogonalization, Different problem of orthogonally and normalization, Tunnelling effect, Characteristics of wave functions, Numerical
Unit 3 Rotational and vibrational motion of a particle
Particle on a sphere, Normalization of the wave functions, Rotation of a diatomic molecule and problems, One-dimensional harmonic oscillator, Hermit's differential Equation, Recursion formula for the Hermit's, differential equation normalization, Eigen functions, Different problems of Eigen function, Harmonic oscillator, Polynomials of different degree and problems
Unit 4 Hydrogen atom
Hydrogen atom, Laguerre and associated Laguerre, *Problems of* Laguerre and associated Laguerre, Recursion formula for the radial solution, Solving the Θ equation and φ equation, Variation method and its Application to Hydrogen molecule, Numerical
Unit 5 Many-electron atoms & angular momenta

The wave functions of many-electron systems, the He atom; many-electron atoms, Hartree-self consistent field methods, angular momenta in many electron atoms, communication with Hamiltonian, spin-orbit Interaction, energy states of atoms, and term symbols, Different Problems

C. Pedagogy

The topics will be discussed in interactive classroom sessions using classical blackboard teaching to power-point presentations/videos. Respective faculty members will also conduct special interactive problem-solving sessions on a weekly basis to solve their difficulties. Course materials will be provided to the students from various primary and secondary sources of information. Internal tests and quizzes will be conducted as part of continuous evaluation and suggestions will be given to students to improve their performance.

D. Learning Resources

1. Quantum Chemistry: Ira N Levine - PHI Publication.
2. Introductory quantum Chemistry: A K Chandra, TMH Publication.
3. Quantum Chemistry: R K Prasad, New age International Publisher.
4. Quantum Chemistry: John P. Lowe, Academic Press

E. Course Outcomes (COs)

CO1	Understand the fundamentals of quantum chemistry
CO2	To explore applications of quantum chemistry in simple systems
CO3	The solution of the Schrodinger equation for different model systems and their correlation with real systems
CO4	Introduction to new operators such as Hermitian and Hamiltonian and their use in the solution of Hydrogen and Hydrogen like atoms

F. Course Articulation Matrix

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**
CO1											
CO2											
CO3											
CO4											

Correlation Levels: 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Course Code: OC762	**Course Name: Stereochemistry and Organic Reactions**
Credit: 4	**Semester: 1**
Contact hours/week	**L** / **T** / **P**: **3** / **1** / **-**

A. Outline of the Course

Sr. No.	**Title of Unit**	**Hours**
1.	**Stereochemistry**	**14**
2.	**Reaction mechanism and methods to determine**	**12**
3.	**Reactive intermediates in organic chemistry**	**14**
4.	**Addition, Elimination and Substitution reactions**	**14**
5.	**Aromaticity**	**06**

B. Detailed Syllabus

Unit 1 Stereochemistry
Optical and geometrical isomerism, Optical activity, chirality and symmetry, Determination of chirality, Different molecular projections and their interconversions, Absolute and relative configuration, Elements of Chirality, Methods of determining configuration, Cahn-Ingold-Prelog system for R/S nomenclature, Optical purity and Enantiomeric excess, Molecules with two (or more) chiral centres and total possible stereoisomers, Compounds containing chiral atom other than carbon, Stereoisomerism in cyclic structures, Prochirality and prostereoisomerism, Homotopic and heterotopic ligands and faces, Stereospecific and stereo-selective reactions
Unit 2 Reaction mechanism and methods to determine
Types of mechanism and reaction, Transition states and intermediates, Thermodynamic and kinetic requirements for reaction, Baldwin rules for ring closure, thermodynamic and kinetic control, Hammond Postulate, Microscopic Reversibility, Marcus theory, Importance of mechanism and methods of determining mechanisms, Determining reaction mechanism through identification of products, Determining reaction mechanism through identification of intermediates, Determining reaction mechanism through isotopic labeling, Determining reaction mechanism through kinetic isotope effects, Determining reaction mechanism through stereochemical evidences, Determining reaction mechanism through cross-over experiment
Unit 3 Reactive intermediates in organic chemistry

Carbocations (carbonium ions and carbenium ions), Structure, geometry and generation of carbocations, Carbocation stability and fate of carbocations, Reactions involving non-classical carbocations, Structure, geometry and generation of carbanions, Stability and reactions of carbanions, Chiral carbanions and tautomerism in carbanions, Structure and generation of free radicals, Stability and reactions of free radicals, Structure, geometry and generation of carbenes, Reactions of carbenes, Structure, reactivity and generation of nitrenes, Reactions of nitrenes, Structure, reactivity, generation and reactions of arynes
Unit 4 Addition, Elimination and Substitution reactions
Addition reactions- orientation, reactivity and mechanism, Markovnikov and anti-Markovnikov addition reactions, Addition reactions (hydro-halo, hydro-hydroxy, and hydro-alkoxy), Addition reactions (dihydro, dihydroxy, dihalo and ozonolysis), Elimination mechanism and orientation, E1-E2-E1cB mechanistic spectrum, Effect of Substrate Structure, attacking base, leaving group and medium on elimination reaction, Pyrolytic Eliminations, Aliphatic substitution, S_E2, S_E1 and S_Ei Mechanism, S_N1, S_N2, S_Ni mechanism, Neighboring-Group mechanism, Aromatic Substitution, S_NAr, S_N1 Mechanism, Orientation and reactivity in mono-substituted benzene rings, Orientation in benzene rings with more than one substituent, Ipso substitution
Unit 5 Aromaticity
Aromaticity and anti-aromaticity, Huckel's rule of aromaticity, Homo-aromaticity (five-, six-, seven- and eight-membered rings), Other systems containing aromatic sextets, Aromaticity of fused rings and Annulenes, Aromatic systems with electron numbers other than six

C. Pedagogy

The topics will be discussed in interactive class room sessions using classical blackboard teaching/molecular models/power-point presentations/video and animations. Weekly tutorial session will be conducted to develop critical thinking and problem solving approach among students. Course materials will be provided to the students from various primary and secondary sources of information. Internal test and quiz will be taken as a part of continuous evaluation and suggestions will be given to student in order to improve their performance.

D. Learning Resources

1. M. B. Smith and J. March, March's Advance Organic Chemistry, 6th Ed. (John Wiley)
2. J. Clayden, N. Greeves, and S. Warren, Organic Chemistry 2nd Ed. (Oxford)
3. Organic Reactions, Stereochemistry and Mechanism: P.S. Kalsi (New Age.)
4. Principles of Organic Synthesis: R.O.C Norman & J.M. Coxon (ELBS)
5. Mechanism in Organic Chemistry: Peter Sykes (Orient Longman)
6. Modern Methods of Organic Synthesis: W. Carruthers (Cambridge)
7. Organic Reaction Mechanism: V. K. Ahluwalia and R. K. Parashar (Narosa)

E. Course Outcomes (COs)

CO1	Visualize spatial arrangement of molecules and understand the stereochemistry of organic molecules and prostereogenic elements

CO2	Understand the generation of reaction intermediates and their reaction pathways.
CO3	Realize the involvement of reactive intermediates in different organic transformations
CO4	Understand the concept of aromaticity and structure-activity relationship of organic molecules.
CO5	Acquaint students about the basic concepts of substitution, elimination, and addition reactions with stereochemical outcomes of products.

F. Course Articulation Matrix

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**
CO1											
CO2											
CO3											
CO4											

Correlation Levels: 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Course Code: OC763	Course Name: Chemical Dynamics and Electrochemistry and Group Theory
Credit: 4	Semester: 1
Contact hours/week	L: 3, T: 1, P: -

L	T	P
3	1	-

A. Outline of the Course

Sr. No.	Title of Unit	Hours
1.	Thermodynamics and Properties of Solution	11
2.	Statistical Thermodynamics	11
3.	Chemical Kinetics	14
4.	Group theory in Chemistry	12
5.	Electrochemistry	12

B. Detailed Syllabus

Unit 1 Thermodynamics and Properties of Solution
Introduction and laws of thermodynamics, Heat engines, Nernst Heat Theorem, Properties of the solution, Raoult's Law, Positive and Negative deviation from ideal behavior, Partial molar properties: partial molar free energy, partial molar volume, partial molar heat content their significances, Numerical, Gibbs Duhem Equation, Density method for Determination of PMV, Numerical
Unit 2 Statistical Thermodynamics
Introduction of statistical thermodynamics, Introduction, the weight of configuration, Numerical of weight configuration, Relation between weight configuration and entropy, Relation between weight configuration and entropy and temperature, Derivation of Equation of translational partition function, Numerical of the translational partition function, Derivation of Equation of Rotational partition function, Numerical of Rational partition function, Derivation of Equation of Vibrational partition function
Unit 3 Chemical Kinetics
Chemical Kinetics and its scope, Rate of reaction, factors influencing the rate of a reaction, measurements of reaction rates, Differential and integral rate laws, rate laws, Integration rate law for nth-order reaction half lifetime, Numerical, Equilibrium constants for elementary reactions, the temperature dependence of rate constants, Arrhenius equation, the concept of activation energy, reaction mechanisms, Examples of activation energy and rate laws, Uni-molecular reactions, bi-molecular reactions, tri-molecular, Uni-molecular reactions, bi-molecular reactions, tri-molecular, Opposing reactions, Consecutive reactions, Parallel reactions, Fast reaction and Equation Derivation, Numerical
Unit 4 Group theory in Chemistry

Introduction of group theory Concepts of symmetry in the molecule:- Symmetry elements, Symmetry operations, definitions and theorems in group theory, Examples of groups, subgroups and classes, Molecular Point groups, Classification, notation of point groups, Different examples of point groups, Matrix for clockwise and anticlockwise Rotation., Matric for the plane of reflection, Matrix representation of point groups: product and square rule, inverse rule, matrices for C_{3v}, $C_{4v,}$ Construction of character tables:-rules, reducible and irreducible representations, character of a representation, Properties of irreducible representations, orthogonally theorem, character tables for C_{2v}, C_{3v}, C_{4v}, D_{nh},
Unit 5 Electrochemistry
Introduction to electrochemistry and conductance, Different Conductometric titration and numerical, DHO equation and its Application, Nernst Equation derivation, Numerical of Nernst equation, Concentration cell with transport, Concentration cell without transport, Numerical of concentration cell, Cyclic voltammetry, Numerical, Demonstration of electrochemical work station instruments, Practical Application of electrochemical work station

C. Pedagogy

The topics will be discussed in interactive classroom sessions using classical blackboard teaching to power-point presentations/videos. Respective faculty members will also conduct special interactive problem-solving sessions on a weekly basis to solve their difficulties. Course materials will be provided to the students from various primary and secondary sources of information. Internal tests and quizzes will be conducted as part of continuous evaluation and suggestions will be given to students to improve their performance

D. Learning Resources

1. Fundamentals of chemical thermodynamics by M.L. Lakhanpal Publisher: Tata McGraw Hill Publishing Company, New Delhi.
2. Thermodynamics by P.C. Rakshit Publisher: The New Book Stall, Calcutta.
3. Principles of physical chemistry by B. R. Puri, L. R. Sharma and M. S. Pathania Publisher: Vishal Publishing Co., Jalandhar, Delhi
4. Physical chemistry by P. W. Atkins and J. de. Paula Publisher: Oxford Press.
5. Statistical thermodynamics by M. C. Gupta Publisher: Wiley-Eastern Ltd. New Delhi.

E. Course Outcomes (COs)

CO1	Understand the fundamentals of thermodynamics, electrochemistry, chemical kinetics, and group theory
CO2	Develop problem-solving ability in electrochemistry, chemical kinetics

CO3	Solve the problem of spectroscopy using the concept of group theory
CO4	Recognize the role of multidisciplinary streams especially basic physics & mathematics science knowledge, in the development of chemistry
CO5	Understood the practical application of chemical Kinetics and electrochemistry.

F. Course Articulation Matrix

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**
CO1											
CO2											
CO3											
CO4											
CO5											

Correlation Levels: 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Course Code: OC764	Course Name: Separation Techniques		
Credit: 4	Semester: 1		
Contact hours/week	L	T	P
	3	1	-

A. Outline of the Course

Sr. No.	Title of Unit	Hours
1.	**Fundamentals of separation techniques**	**10**
2.	**Gas chromatography**	**12**
3.	**High performance liquid chromatography and Ultra performance liquid chromatography**	**14**
4.	**Liquid-liquid Partition chromatographic techniques**	**14**
5.	**Other chromatographic techniques**	**10**

B. Detailed Syllabus

Unit 1 Fundamentals of separation techniques
Classification of separation techniques, Principle of solvent extraction, Method of separating mixture, Introduction of solvent extraction, Principle of solvent extraction, Distribution coefficient. Advantages and disadvantages of solvent extraction, Solvent extraction techniques**:** Batch extraction, Counter current extraction, Retention time, Retention factor, Height Equivalent to Theoretical plate (HETP), column efficiency, Resolution, Zone broadening, Van deemter's Equation, Discussion of various term in Van deemter's Equation
Unit 2 Gas chromatography
Fundamentals of Gas chromatography, Instrumentation of GC, Stationary phase , carrier gas, Key operation variables controlling retention, Different types of column, Injection port, column component, Detector specification – Flame ionization detector, Thermal conductivity detector, Electron capture detector, Gas density balance detector, Application in the analysis of gases, organic compounds, Determination of Butyl lithium and volatile halogenated hydrocarbons, Determination of carbon in H_2O_2, Isolation and identification of light oil alkanes in shale oil, To determine heat of evaporation of aromatic hydrocarbons, Identification of the components of hetrologous series – Using retention data and functionality test, Determination of empirical formula
Unit 3 High performance liquid chromatography and Ultra performance liquid chromatography
Fundamental of liquid Chromatography separation, Instrumentation, Various Column and guard column, Normal phase and reverse phase chromatography, Isocratic and gradient illusion, Retention factor, Peak Symmetry: Asymmetry Factor and Tailing Factor, Factors affecting column efficiency, UV detector, RI detector, Photo Diode Array detector, ELSD detector, Advantages and disadvantages, Introduction of UPLC, Advantages and disadvantages of UPLC
Unit 4 Liquid-liquid Partition chromatographic techniques

Flash Chromatography: Introduction of flash chromatography, Instrumentation of flash chromatography, Packing of column, selection of solvent, Super critical fluid chromatography: Instrumentation of SFC, stationary and mobile phases used in SFC, Advantages and Application of SFC, Gel permeation chromatography: Instrumentation, Retention behaviour, resolution, Application of GPC, Ion exchange chromatography: Ion exchange resins, structure of resins, Ion exclusion, ligand exchange, Affinity chromatography: Principle and instrumentation, Chromatographic Media, Immobilized Ligand, Advantages and disadvantages of Affinity chromatography

Unit 5 Other chromatographic techniques

Thin Layer Chromatography: Introduction and principle, chromatographic media-coating materials, Activation of adsorbent, Sample development, Development of chromatoplate, solvent systems, Types of development, visualization methods, HPTLC: Introduction, sample application, Micro – preparative chromatography, Quantitative evaluation, Application of HPTLC

C. Pedagogy

The topics will be discussed in interactive class room sessions using classical blackboard teaching to power-point presentations. Special interactive session in the form of seminars will be also conducted by respective faculty members on weekly bases. Each student will give one seminar for a course. Course materials will be provided to the students from various primary and secondary sources of information. Internal test and quiz will be conducted as a part of continuous evaluation and suggestions will be given to students in order to improve their performance.

D. Learning Resources

1. Fundamentals of analytical Chemistry by D. A. Skoog, D. M. West, F. J. Holler, S.R. Crouch, Ninth edition. (CENGAGE Learning)
2. Basic Gas Chromatography by McNiar. (Wiley)
3. HPLC Edited by Yuri Kazakevich & Rosario Lobratto. (Wiley)
4. Separation methods by M N Sastri. (*Himalaya Publishing Company*)
5. Analytical Chemistry by *Gary Christian, seventh edition. (Wiley)*
6. *Analytical Chemistry: an Introduction by* Douglas A. Skoog, Donald M. West and F. James Holler, fifth edition, Saunders college publishing, New York.
7. Analytical Chromatography by Gurdeep R Chatwal. (*Himalaya Publishing Company*)
8. Instrumental method of analysis, H. H. Willard, seventh edition. (Wadsworth publishing company)

E. Course Outcomes (COs)

CO1	Understanding the basic principle of various separation techniques.
CO2	To know about use of analytical techniques like GC, HPLC, HPTLC and other separation techniques in sample analysis.
CO3	To study importance of analytical techniques and its interdisciplinary applications.
CO4	Apply the analytical techniques for qualitative and quantitative analysis of unknown samples

F. Course Articulation Matrix

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**
CO1											
CO2											
CO3											
CO4											

Correlation Levels: 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Course Code: OC765	Course Name: Organic Chemistry Laboratory I
Credit: 2	Semester: 1
Contact hours/week	L: - , T: - , P: 4

L	T	P
-	-	4

A. List of Experiments

Sr. No.	List of Experiments
1.	**Introduction, Record Keeping, and Laboratory Safety** • Introduction • Preparing for the Laboratory • Working in the Laboratory • The Laboratory Notebook • General Protocol for the Laboratory Notebook • Types of Organic Experiments and Notebook Formats • Sample Calculations for Notebook Records • Safe Laboratory Practice: Overview • Safety: General Discussion • Safety: Material Safety Data Sheets (MSDSs) • Safety: Disposal of Chemicals 2
2.	**Techniques and Apparatus** • Glass wares & Assembling Apparatus • Weighing Methods • Filtration Apparatus and Techniques • Solids: Recrystallization and Melting Points • Liquids: Distillation and Boiling Points
3.	**Thin Layer Chromatography** • Introduction & Theory

	• Determination of R_f Values
4.	**Preparations:** • Dibenzalacetone from Benzaldehyde (Claisen – Schmidt Condensation) • Methyl Orange from Sulphanilic acid (Diazotization & Coupling) • 9,10-dihydro anthracene-endo-α,β-succinic anhydride from Anthracene & Maleic anhydride (Diel's Alder Reaction) • p-iodonitrobenzene from p-nitroaniline (Sandmayer Reaction) • P-aminobenzoic acid to p-cholorobenzoic acid (Sandmayer Reaction) • Cyclohexanone to adipic acid (Oxidation) • p-chlorobenzaldehyde to p-chlorobenzoic acid and p-chlorobenzyl alcohol (Cannizzaro Reaction)
5.	**Estimations:** • Determine the amount of Phenol in the given solution. • Determine the amount of Crotonic acid in the given solution.
6.	**Demonstrative Practicals:** • Determination of λ_{max} of the synthesized compound using UV-Visible spectroscopy. • Determination of the presence of functional group using Infra-red Spectroscopy.

B. Pedagogy

The course will give emphasis on active learning in students through utilizing the theoretical knowledge into the practical knowledge**.** Demonstrative Practicals is helpful to the students to learn the spectroscopic techniques. Students will be continuously evaluated through viva-voce for each experiment and each student will appear in internal test.

C. Learning Resources

1. Elementary practical of Organic Chemistry: A.I. Vogel (Longman group Ltd.)

2. College Practical Chemistry: V K Ahluwalia & Sunita Dhingra (Universities Press (India) Pvt. Ltd.) 3. Practical Organic Chemistry: F. G. Mann & B. C. Saunders (Pearson Education) 4. Experimental Organic Chemistry: John Gilbert & Stephen Martin 6th Ed. (Cengage learning). 5. Advanced Practical Organic Chemistry: J. Leonard, B. Lygo and G. Procter (CRC Press).

D. Course Outcomes (COs)

CO1	Students will learn how to maintain laboratory safety
CO2	They will know the use of different methods for synthesizing the organic compounds.
CO3	They will be prepared for next higher level experiments.

E. Course Articulation Matrix

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**
CO1											
CO2											
CO3											

Correlation Levels: 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Course Code: OC766	Course Name: Inorganic Chemistry Laboratory-I
Credit: 2	**Semester: 1**
Contact hours/week	**L** / **T** / **P**: - / - / 4

A. List of Experiments

1.	Semi-micro qualitative analysis of inorganic mixture (6 + 1 radicals) [max. 6 practical] 6 – Cation / Anion (variable) 1 – Rare earth element from the following Th, Li, Ce, Mo, Te, Se, V, Ti, and Zr
2.	FTIR-sample preparation and data collection (Demonstrative Practical)
3.	Identification of few inorganic compounds through analysis of FTIR spectra

B. Learning Resources

1. Qualitative Inorganic Analysis– A. I. Vogel, 6th Edition revised by G. Svehla ELBS–London
2. Textbook of Chemistry Analysis – A. I. Vogel
3. Qualitative Chemistry semi micro analysis – edited by P. K. Agasyan CBS Publisher-Delhi.
4. Chemistry: Inorganic Qualitative Analysis in the Laboratory, Clyde Metz, Elsevier, 2012.

C. Pedagogy

Hands on experience for the semi-micro qualitative analysis of inorganic mixture through systematic chemical tests on the given mixture. Students will learn sample preparation and data collection using FTIR spectrophotometer. Analysis of FTIR spectra of some inorganic compounds. Internal test and viva will be taken as a part of continuous evaluation.

D. Course Outcomes (COs)

CO1	Understand the fundamentals of qualitative analysis.
CO2	Gain knowledge about the identification of radicals in the inorganic mixture.
CO3	Able to analyze the presence of functional groups through FTIR spectroscopy.
CO4	Acquaint students with semi-micro analysis method.

E. Course Articulation Matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1											
CO2											
CO3											
CO4											

Correlation Levels: 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Course Code: OC767	Course Name: Physical Chemistry Laboratory-I
Credit: 2	Semester: 1
Contact hours/week	L: - / T: - / P: 4

L	T	P
-	-	4

A. List of Experiments

Sr. No.	Experiments
1.	(1) To find out the cell constant of a given conductivity cell (2) To determine the critical micelle concentration of anionic surfactant. **(Conductometer CMC)**
2.	***To verify the additivities of absorbance of a mixture of colored substances in solution using potassium permanganate and potassium dichromate solution.*** ***(Spectrophotometer)***
3.	etermination of concentration/ amounts of iodide, bromide and chloride in the mixture by potentiometric titration with silver nitrate. **(Potentiometer)**
4.	etermination of ΔG, ΔH, and ΔS for a reaction using an electrochemical cell. **(Potentiometer)**
5.	To determine the dissociation constants (k1 and k2) for a dibasic acid. **(pH Meter)**
6.	To find the unknown concentration of sugar solution using a polarimeter. (**Polarimeter**)
7.	**To determine the rate of acid-catalyst iodination of acetone in the presence of Excess acid & acetone at room temperature. (Chemical Kinetics)**
8.	To determine the composition of a binary liquid mixture by refractometry. (**Refractometer**)
9.	Determine the surface tension of liquids (methyl acetate, ethyl acetate, Chloroform, hexane ....) and hence calculate the atomic parachors of C, H & Cl. **(Surface Tension)**
10.	**To determine Partial Molal Volumes of Sodium Chloride in an aqueous solution at room temperature. (Partial Molal Volume)**

C. Pedagogy

The students will be taught how to do experiments with instruments like pH meter, conductometer, polarimeter, potentiometer, Refractometer, and spectrophotometer for exercises in the physical chemistry laboratory. They will also perform non-instrumental experiments to understand the other concept of physical chemistry. Internal tests and viva will be taken as a part of a continuous evaluation

D. Learning Resources

1. Experimental Physical Chemistry by R. C. Das & B. Behera, (Tata McGraw hill Publishing Company Ltd., New Delhi)
2. Systematic Experimental Physical Chemistry by S. W. Rajbhoj and T. K. Chondhekar, (Anjali Publication, Aurangabad)
3. Advanced Practical Physical Chemistry by J. B. Yadav, (Goel Publishing House, Meerut)
4. Experimental Physical Chemistry by G. Peter Matthews, (Clarendon Press, Oxford, London)
5. Experimental Physical Chemistry by V. D. Athawale and ParulMathur, (New Age International Publishers, New Delhi)
6. Advanced Physical Chemistry Experiments by Gurtu and Gurtu, (Pragati Prakashan, Meerut)

E. Course Outcomes (COs)

CO1	Acquire knowledge about different methods of preparation of the solution.
CO2	Students learn how to take measurements with instruments like Refractometer, Conductometer, pH Meter, and Spectrophotometer and interpret the results correctly.
CO3	Understand the concept of chemical Kinetics and thermodynamics through experiments.
CO4	Develop the skill of interpreting data from various experiments, including the construction of appropriate graphs and evaluation of errors.

F. Course Articulation Matrix

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**
CO1											
CO2											
CO3											
CO4											

Correlation Levels: 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

HS105.02 E: ACADEMIC SPEAKING AND PRESENTATION SKILL (Sem.-I)

1. Credits and Schemes:

Sem.	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
1	HS704 A/B/C/ D/E/F/ G/H	Academic speaking and presentation skill	02	02	--	--	30	70	100

2. Course Outline

Module No.	Title/Topic	Classroom Contact Hours
1	**Foundations of Advance Communication** • *Meaning and Definition of Advance Communication* • *Advance Communication in Digital, Social, Mobile World* • *Strategies for Advance Communication* • Meaning and Concept of Academic Language • High Frequency Academic Vocabulary	**04**
2	**Art of Conversation** • *Describing people, places and things* • *Expressing opinions* • *Making suggesting* • *Persuading someone* • *Interpreting and Summarizing*	**06**
3	***Science of Power Speaking*** • *Phonemes* • *Word Stress*	**06**

	• *Pronunciation* • *Intonation* • *Pause* • *Register* • *Fluency* • *Prosody* • *Lexical Range*	
4	***Academic Speaking Application – Part I*** • *Art of Oratory* • *Formal Presentation* • *Speech Analysis – Decoding Best Speeches*	**08**
5	**Academic Speaking Application – Part II** • *Job Interview* • *Group Discussion* • *Meeting*	**06**
	Total	**30**

Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like case studies, group work, independent and collaborative research, presentations etc.

I. Evaluation

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Numbe r	Marks per incidence	Total Marks
1	I-Talk	1	10	25
2	Situational Speaking	1	05	
3	Case Study - Speech Analysis	2	10	
4	Attendance and Class Participation	-		05
			Total	**30**

External Evaluation

The University Practical Examination will be for 70 marks and will test the advance communication skills and academic speaking.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical	-	70	70
			Total	**70**

Course Outcome (COs):

After completion of the course the student would :

CO1	understand and demonstrate advance communication skills and academic speaking.
CO2	demonstrate linguistic competence
CO3	demonstrate performing ability at group discussion and personal interview.
CO4	demonstrate the formal presentation skills.
CO5	demonstrate ability to communicate in diverse situations

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	-	-	3	-	-	-
CO2	-	-	-	-	-	-	-	3	-	-	-
CO3	-	-	-	-	-	-	-	3	-	-	2
CO4	-	-	-	-	-	-	-	3	-	-	-
CO5	-	-	-	-	-	-	-	3	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Reference Books

- Headway Academic Skills - Level 1: Listening, Speaking and Study Skills Student's Book Paperback

Reading

- **Unit 1:** Business communication Today (Thirteenth Edition) by Courtland L. Bovee, John V. Thill and Roshan Lal Raina
- **Unit 2:** Effective Speaking Skills by Terry O' Brien
- **Unit 2:** Speak Better Write Better by Norman Lewis
- **Unit 2:** Well Spoken: Teaching Speaking to All Students by Erik Palmer
- **Unit 3:** Let Us Hear Them Speak : Developing Speaking – Listening Skills in English by Jayshree Mohanraj (Publisher – Sage Publication)
- **Unit 4:** The craft of scientific presentations: Critical steps to succeed and critical errors to avoid. New York: Springer by Michael Alley
- **Unit 4:** Presentation Skills in English by Bob Dignen (Publisher: Orient Black Swan)

Course Code: OC771	Course Name: Spectroscopy I		
Credit: 4	Semester: 2		
Contact hours/week	L	T	P
	3	1	-

A. Outline of the Course

Sr. No.	Title of Unit	Hours
1.	**Molecular Formulas and What can be learned from them**	05
2.	**UV-Visible Spectroscopy**	15
3.	**FTIR Spectroscopy**	15
4.	**Mass Spectroscopy**	15
5.	**Microscopic Techniques**	10

B. Detailed Syllabus

Unit 1 Molecular Formulas and What can be learned from them

- Elemental Analysis and Calculations.
- Determination of Molecular Mass.
- Molecular Formulas.
- Index of Hydrogen Deficiency.
- The Rule of Thirteen.

Unit 2 UV-Visible Spectroscopy

- Introduction.
- Laws of Absorption (Lambert-Beer's Law).
- Theory of Electronic transitions and Ultraviolet absorption.
- Chromophores and Auxochromes (Hyper, Hypo, Batho and Hypso Chromic effect).
- Woodward-Fieser rules for Dienes.

- Fieser-Kuhn rule for highly conjugated dienes.
- Woodward-Fieser rules for enones.
- Characteristic absorptions in alkenes and alkynes alcohols, ethers, amines, carbonyl compounds.
- Characteristic absorptions in aromatic compounds.
- λmax Calculation of Aromatic Compounds.
- Factors influencing λmax.
- Effects of conjugation and the effect of solvent.
- Differentiation of compounds / isomers by UV.

Unit 3 FTIR Spectroscopy

- Introduction.
- Theory and principles.
- Molecular vibrations.
- Calculations of vibrational frequencies.
- Factors influencing IR frequency.
- Characteristic group absorptions in hydrocarbons, aromatic compounds, alcohol and phenols, ethers, carbonyl compounds, amines, nitriles, nitro compounds, carboxylic acids and halide.
- Differentiation of compounds/isomers by IR.

Unit 4 Mass Spectroscopy

- Introduction.
- Theory and principles of mass spectroscopy.
- Instrumentation.
- Low and High resolution mass spectra.
- Ionization techniques – Electron Impact (EI) ionization and Chemical Ionization (CI).
- Determination of molecular weight and molecular formula.
- Nitrogen rule.

- Detection of molecular ion peak, metastable ion peak.
- Fragmentations – rules governing the fragmentations.
- Mc Lafferty rearrangement.
- Interpretation of mass spectra of different class of compounds – saturated and unsaturated hydrocarbons, aromatic hydrocarbons, alcohols, ethers, ketones, aldehydes, carboxylic acids, amines, amides, compounds containing halogens.

Unit 5 Microscopic Techniques

- Introduction
- Scanning Electron Microscopy (SEM) – Basic Principle, Instrumentation, Operating parameters and Applications
- Scanning Tunneling Microscopy (STM) – Basic Principle, Instrumentation, Operating parameters and Applications
- Atomic Force Microscopy (AFM) – Basic Principle, Instrumentation, Operating parameters and Applications

C. Pedagogy

The topics will be discussed in interactive classroom sessions using classical blackboard teaching/molecular models/power-point presentations/videos and animations. The weekly tutorial session will be conducted to develop students' critical thinking and problem-solving approach. Course materials will be provided to the students from various primary and secondary sources of information. Internal tests and quizzes will be taken as a part of continuous evaluation, and suggestions will be given to the student to improve their performance.

D. Learning Resources

1. Spectrometric Identification of Organic Compounds: Robert M. Silverstein & Francis X. Webster 7th Ed. (Wiley).

2. Introduction to Spectroscopy: Donald L. Pavia, Gary M. Lampman, George S. Kriz & James R. Vyvyan 5th Ed. (Cengage Learning).

3. Spectroscopy of Organic Compounds: P. S. Kalsi 6th Ed. (New Age International Publishers). **4.** Organic Spectroscopy: Willliam Kemp 3rd Ed. (Palgrave Macmillan) **5.** Fundamentals of Molecular Spectroscopy: Colin N. Banwell & Elaine M. McCash 4th Ed. (McGrawHill Education). **6.** Principles of Instrumental Analysis: Douglas A. Skoog, F. James Holler & Stanley R. Crouch 7th Ed. (Cengage Learning).

E. Course Outcomes (COs)

CO1	The basic principles involved in the interaction of electromagnetic radiation with matter.
CO2	Gain insight into the basic principles of UV-Visible, FTIR and Mass spectroscopic techniques.
CO3	Understand basic components of UV-Visible, FTIR and Mass Spectrometer.
CO4	Interpretation of UV-Visible, FTIR and Mass Spectra.

F. Course Articulation Matrix

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**
CO1											
CO2											
CO3											
CO4											

Correlation Levels: 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Course Code: OC772	Course Name: Name Reactions and Reagents in Organic Chemistry		
Credit: 4	Semester: 2		
Contact hours/week	L	T	P
	3	1	-

A. Outline of the Course

Sr. No.	Title of Unit	Hours
1.	Organic name reactions, mechanism and application-I	15
2.	Organic name reactions, mechanism and application-II	15
3.	Reagents in organic synthesis-I	15
4.	Reagents in organic synthesis-II	15

B. Detailed Syllabus

Unit 1 Organic name reactions, mechanism and application-I
Robinson Annulation and related reactions, Importance of Mannich reaction in Robinson Annulation, Stork Enamine reaction, Wittig reaction and its modification, Peterson olefination, Shapiro and Bamford Steven's reaction, Julia Olefination, Vilsmayer-Heck reaction, Mitsonobu reaction, Buchwald-Hartwing amination, Brown's Hydroboration, Hydroboration-oxidation reaction, Applications of hindered boron reagents, Carbonylation of organoboranes, Stobbe condensation
Unit 2 Organic name reactions, mechanism and application-II
Stille coupling, Suzuki coupling, Sonogashira coupling, Heck reaction, Negishi coupling, Mukaiyama esterification, Ritter reaction, Yamaguchi esterification, Domino reactions and its characteristics, Nazerov cyclization, Ugi reaction, Strecker synthesis, Click reactions and its characteristics, Huisgen 1,3-dipolar cycloaddition
Unit 3 Reagents in organic synthesis-I
DCC, LDA, Grignard reagent, Gilman reagent, Alkyl lithium, 1,3-Dithianes, CrO_3, MnO_2, SeO_2, $KMnO_4$, $Pb(OAc)_4$, OsO_4, HIO_4, DMSO, H_2O_2
Unit 4 Reagents in organic synthesis-II
Ozone, HgO, NBS, $K_3Fe(CN)_6$, DDQ, $Al(O\text{-}t\text{-}Bu)_3$, $Al(O\text{-}iPr)_3$, Dess-Martin, $LiAlH_4$, $NaBH_4$, Complex hydrides, DIBAL, Zn-Hg/HCl, $N_2H_4/^{-}OH$, TBTH, Cat. H_2, Wilkinson catalyst

C. Pedagogy

The topics will be discussed in interactive class room sessions using classical blackboard teaching/molecular models/power-point presentations/video and animations. Weekly tutorial session will be conducted to develop critical thinking and problem solving approach among students. Course materials will be provided to the students from various primary and secondary sources of information. Internal test and quiz will be taken as a part of continuous evaluation and suggestions will be given to student in order to improve their performance.

D. Learning Resources

1. M. B. Smith and J. March, March's Advance Organic Chemistry, 6th Ed. (John Wiley)
2. J. Clayden, N. Greeves, and S. Warren, Organic Chemistry 2nd Ed. (Oxford)
3. Advanced Organic Chemistry: Reaction Mechanisms, R. Bruckner, Academic Press (2002).
4. Principles of Organic Synthesis: R.O.C Norman & J.M. Coxon (ELBS)
5. Organic reactions & their mechanisms, third revised edition, P.S. Kalsi, New Age International Publishers.
6. Modern Methods of Organic Synthesis: W. Carruthers (Cambridge)
7. Organic Reaction Mechanism: V. K. Ahluwalia and R. K. Parashar (Narosa)

E. Course Outcomes (COs)

CO1	Recognize the reaction mechanism and reaction pathways.
CO2	Realize the uses of various types of oxidants and reducing agents in organic synthesis.
CO3	Understand the importance of name reactions and molecular rearrangements.
CO4	Gain knowledge about the industrial applications of name reactions and reagents.
CO5	Acquaint students about the different synthetic strategies.

F. Course Articulation Matrix

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**
CO1											
CO2											
CO3											
CO4											

Correlation Levels: 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Course Code: OC773	**Course Name: Topics in Physical Chemistry**
Credit: 4	**Semester: 2**
Contact hours/week	**L** / **T** / **P**: **3** / **1** / **-**

A. Outline of the Course

Sr. No.	Title of Unit	Hours
1.	**Fugacity and Partial Molar Properties**	13
2.	**Molecular weight determination of Polymers and Kinetics**	13
3.	**Introduction to Nanomaterials**	10
4.	**Surface Tension & Intermolecular Forces**	12
5.	**Solid State Chemistry**	12

B. Detailed Syllabus

Unit 1 Fugacity and Partial Molar Properties

Introduction to Fugacity, Methods for Determination of Fugacity, Relative fugacity, Different Methods for Determination of fugacity for pure gas, Graphical and approximate method, Equation of state method, Numerical for Determination of fugacity, Determination of fugacity of the mixture of gas, Clapeyron - Clausius equation, Partial Molar Properties, Method for Determination of Partial Molar Properties, Numerical

Unit 2 Molecular weight determination of Polymers and Kinetics

Introduction of polymer chemistry, Classification of Mol. Wt. determination methods
Based on colligative properties, End group analysis method, Numerical elevation of boiling point and depression in freezing point method, Membrane osmometry and Vapour pressure osmometry method, Derivation of Equation of Numerical, Viscosity, GPC, and light scattering method, Mechanism of Free radical polymerization, Kinetics of free radical polymerization, Kinetics of copolymerization, Reactivity ratio

Unit 3 Introduction to Nanomaterials

Introduction, Fundamentals, Properties and Size dependence of properties, Chemical Optical, vibrational, thermal Electrical Magnetic Mechanical, Nano Material characterization techniques, Applications

Unit 4 Surface Tension & Intermolecular Forces

Introduction of surface phenomena, *Methods for Determination of Surface tension method capillary rise and numerical, Methods for Determination of Surface tension method Drop weight and drop number, numerical,* Surface active agents, Classification of surface-active agents, Micellization, hydrophobic Interaction, Critical micelle concentration (CMC), factors affecting the CMC of surfactants Different methods for Determination of CMC, Determination of surface parameter, Dipole moments & polar molecules Intermolecular forces between molecules, *Numerical*

Unit 5 Solid State Chemistry

Introduction and different cubic structures, Calculation of coordination number, packing fraction and relation between a and r for SC, Calculation of coordination number, Packing fraction and relation between a and r for BCC, Calculation of coordination number, Packing fraction and relation between a and r for FCC, Miller indices and examples, Bragg Equation. X-ray diffraction pattern of the cubic system, Different numerical of Braggs laws, Crystal defects Vacancy defect, Interstitial defect, Impurity defects

C. Pedagogy

The topics will be discussed in interactive classroom sessions using classical blackboard teaching to power-point presentations/videos. Respective faculty members will also conduct special interactive problem-solving sessions on a weekly basis to solve their difficulties. Course materials will be provided to the students from various primary and secondary sources of information. Internal tests and quizzes will be conducted as part of continuous evaluation and suggestions will be given to students to improve their performance.

D. Learning Resources

1. Physical Chemistry by P.W. Atkins and J.de Paula, Oxford Press.
2. Principles of Physical Chemistry by B. R. Puri, L.R. Sharma and M. S. Pathania ; Vishal Publishing House.
3. Physical Chemistry by G. M. Barrow, McGraw-Hill
4. Physical Chemistry of Polymers by A. Tagger , Mir Publisher, Moscow ,1980. 39
5. Text Book of Polymer Science by PadmaL. Nayak and S. Lenka, Kalyani Publisher, New Delhi.
6. Text Book of Polymer Science by Fred W. Billmeyer, John Wiley and Sons.
7. The Physics and Chemistry of Nano Solids by Frank J. Owens and Charles P. Poole Jr, Wiley-Interscience, 2008.
8. Introduction to solids by L. V. Azaroff, McGraw-Hill.
9. Solid State Chemistry and its Applications by A. R. West, Wiley 10. Principles of the solid state by H. V. Keer, Wiley –Eastern Ltd.

E. Course Outcomes (COs)

CO1	Develop the understanding of the fundamental concept of surface chemistry, nanomaterials and solid state
CO2	Apply the knowledge the surface chemistry to the emerging problem in basic science
CO3	Understood the practical Application of polymer chemistry, solid state and nanomaterials
CO4	Develop problem-solving ability in the field of solid state and nanomaterials

F. Course Articulation Matrix

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**
CO1											
CO2											
CO3											
CO4											

Correlation Levels: 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Course Code: OC774	**Course Name: Coordination chemistry and magneto Chemistry**
Credit: 4	**Semester: 2**
Contact hours/week	**L**: 3, **T**: 1, **P**: -

L	T	P
3	1	-

A. Outline of the Course

Sr. No.	Title of Unit	Hours
1.	**Coordination Chemistry: I**	12
2.	**Coordination Chemistry: II**	12
3.	**Bonding in Transition Metal Complexes**	12
4.	**Magnetochemistry I**	12
5.	**Magnetochemistry: II**	12

B. Detailed Syllabus

Unit 1 Coordination Chemistry: I

Fundamentals of coordination, Determination of shape of the d orbitals using different mathematical Equation, Chemistry Concept of crystal field theory (CFT), Ligand field theory (LFT), Splitting of d-orbitals in various octahedral, Tetragonal, elongation and compression in octahedral complexes, John teller distortion, Numerical

Unit 2 Coordination Chemistry: II

Nephelauxetic series, Derivation of terms for closed subshell, Derivation of terms for D, Derivation of terms for f, Configurations, Correlation diagrams for octahedral, tetrahedral system Numerical, Configurations, Correlation diagrams for octahedral

Unit 3 Bonding in Transition Metal Complexes

Tanabe-Sugano diagram for octahedral and tetrahedral stereochemistry of d1 to d9 systems, Determination of Crystal Field Splitting Energy and electronic parameters for octahedral and tetrahedral complexes, Charge Transfer transitions. Molecular orbital energy level diagram for the octahedral and tetrahedral complexes

Unit 4 Magnetochemistry I

Magnetic susceptibility: Sources of paramagnetism, diamagnetic susceptibility, Pascal constants and constitutive corrections Langevin equation, Van Vleck's formula, antiferromagnetism, Types of antiferromagnetism, antiferromagnetic exchange pathways, Ferromagnetism and magnetic domains, molecular field theory of ferromagnetism, magnetic sub lattice, ferrimagnetism and canting.
Unit 5 Magnetochemistry: II
Spin-orbit coupling, Lande's interval rule, quenching of orbital magnetic moment by crystal field, spin-orbit coupling on A and E terms, spin-orbit coupling on T term, Spin paring: spin paring in octahedral complexes; spin paring in non-octahedral complexes; some aspects of spin pairing and cross over the region.

C. Pedagogy

The topics will be discussed in interactive classroom sessions using classical blackboard teaching to power-point presentations/videos. Respective faculty members will also conduct special interactive problem-solving sessions on a weekly basis to solve their difficulties. Course materials will be provided to the students from various primary and secondary sources of information. Internal tests and quizzes will be conducted as part of continuous evaluation and suggestions will be given to students to improve their performance.

D. Learning Resources

1. Elements of Magnetochemistry by R. L. Dutta and A. Syamal, Affiliated East-West Press Pvt. Ltd. New Delhi.
2. Introduction to Ligand Fields by B. N. Figgis, Inter science, New York.
3. Physical Methods in Chemistry by R. S. Drago, W. B. Saunders.
4. Inorganic Electronic Spectroscopy by A. B. P. Lever ; Elsevier, Amsterdam.
5. Magnetism and Transition Metal Complexes by F E. Mabbs and D. J. Machin, Chapman and Hall, London.
6. Electronic Absorption Spectroscopy and Related Techniques by D. N. Sathyanarayana; Universities Press, Hyderabad.

E. Course Outcomes (COs)

CO1	Understand the basics of coordination chemistry including coordination numbers and geometry
CO2	Ensuring that students acquire sound knowledge to value the importance of electronic transitions and magnetic property
CO3	Interpret the electronic spectra of coordination compounds explaining color, allowed and forbidden transitions through Orgel and Tanabe-Sugano diagrams

CO4	Demonstrate the basics, spectral and magnetic properties of Lanthanides and Actinides

F. Course Articulation Matrix

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**
CO1											
CO2											
CO3											
CO4											

Correlation Levels: 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Course Code: OC775	Course Name: Organic Chemistry Laboratory II
Credit: 2	**Semester: 2**
Contact hours/week	L: - / T: - / P: 4

L	T	P
-	-	4

A. List of Experiments

Sr. No.	List of Experiments
1.	**Use of Computer – Chem Draw, Chem-Sketch and ISI-Draw** • Draw the Structure of Simple aliphatic, aromatic and heterocyclic compounds with different substituents. • IUPAC Names • Predict the ^{1}H NMR and ^{13}C NMR data from the software.
2.	**Preparations:** • 2,4-Dihydroxybenzoic acid from Resorcinol (Oxidation) • m-nitroaniline from m-dinitrobenzene (Reduction) • 2,5-dimethylbenzenesulfonic acid. (Sulphonation) • p-nitroanilne from acetanilide (Nitration) (2-step) • Picric acid from Phenol (Nitration) • p-bromonitrobenzene from p-bromo benzene (Nitration) • p-bromoaniline from Acetanilide (Bromination) (2-Step) • 2,4,6-tribromophenol from Phenol (Bromination) • Acetanilide from Acetophenone (Beckman Rearrangement) (2-Step) • Benzilic Acid from Banzoin (Benzil-Benzilic acid rearrangement) (2-Step)
3.	**Estimation:** • Determine the amount of Aniline in the given solution. • To determine percentage purity of Carbonyl compound (Aldehydes & Ketones).
4.	**Demonstrative Practicals:**

	• Determination of λ_{max} of the synthesized compound using UV-Visible spectroscopy. • Determination of the presence of functional group using Infra-red Spectroscopy.

B. Pedagogy

The course will give emphasis on active learning in students through utilizing the theoretical knowledge into the practical knowledge**.** Demonstrative Practicals is helpful to the students to learn the spectroscopic techniques. Students will be continuously evaluated through viva-voce for each experiment and each student will appear in internal test

C. Learning Resources

1. Elementary practical of Organic Chemistry: A.I. Vogel (Longman group Ltd.)
2. College Practical Chemistry: V K Ahluwalia & Sunita Dhingra (Universities Press (India) Pvt. Ltd.)
3. Practical Organic Chemistry: F. G. Mann & B. C. Saunders (Pearson Education)
4. Experimental Organic Chemistry: John Gilbert & Stephen Martin 6th Ed. (Cengage learning).
5. Advanced Practical Organic Chemistry: J. Leonard, B. Lygo and G. Procter (CRC Press).

D. Course Outcomes (COs)

CO1	Students will learn how to draw and predict the spectroscopic data with help of computer.
CO2	They will know the use of different methods for synthesizing the organic compounds.
CO3	They will be prepared for next higher level experiments.

E. Course Articulation Matrix

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**
CO1											
CO2											
CO3											

Correlation Levels: 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Course Code: OC776	Course Name: Inorganic Chemistry Laboratory-II		
Credit: 2	**Semester: 2**		
Contact hours/week	**L**	**T**	**P**
	-	-	4

A. List of Experiments

1.	Synthesis of metal complexes and double salts [Any Four]
	• Hexaminenickel (II) chloride • Tris(thiourea)copper(I) sulphate • Ferrous ammonium sulphate • Ammonium copper(ll) sulphate • Tris (acetylacetonato) iron (III) • Cis- bis (glycinato) copper (II) monohydrates • Any other preparation in line with above-mentioned
2.	Quantitative analysis of some synthesized metal complexes and double salts by gravimetric and volumetric methods
3.	Synthesis of nanomaterials [Any Two]
	• Synthesis of metal oxide nanoparticle (Fe_3O_4/ZnO/TiO_2) • Synthesis of polymeric nanocomposite (PbS-PVA/PVP) • Any other preparation in line with above-mentioned
4.	Characterization of synthesized materials by UV-Vis, FTIR, PXRD and TGA (Demonstrative Practical)

B. Learning Resources

1. Mendham, J., Denney, R. C., Barnes, J. D., Thomas, M. and Sivasankar, B. Vogel's Quantitative Chemical Analysis, 6th Edn., (Pearson Education, 2009).
2. Marr, G., Rockett, B. W. Practical Inorganic Chemistry, (Van Nostrand, 1972).
3. Wollins, J. D. Inorganic Experiments, 3rd Edn., (VCH, 1994).
4. Parshall, G. W. (Ed. in Chief). Inorganic Synthesis, Vol. 15, (McGraw Hill, 1974).

C. Pedagogy

Hands on experience for the synthesis of metal complexes, double salts and nanomaterials. Students will learn quantitative analysis by gravimetric and volumetric methods. Training of sophisticated instruments such as UV-Vis, FTIR, PXRD and TGA for the characterization of materials. Internal test and viva will be taken as a part of continuous evaluation.

D. Course Outcomes (COs)

CO1	Understand the chemistry of metal complexes.
CO2	Familiarize with different synthetic approaches to synthesize inorganic materials.
CO3	Able to recognize the right analytical tool for the characterization of inorganic materials.
CO4	Understand the different aspects of quantitative analysis.

E. Course Articulation Matrix

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**
CO1											
CO2											
CO3											
CO4											

Correlation Levels: 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

Course Code: OC777	Course Name: Physical Chemistry Laboratory-II		
Credit: 2	**Semester: 2**		
Contact hours/week	**L**	**T**	**P**
	-	-	4

A. List of Experiments

Sr. No.	Experiments
1.	***To determine the molecular composition of ferric – salicylate complex by Job's method. (Spectrophotometer)***
2.	determine the cell constant of a given conductance cell and the equivalent conductance of a strong electrolyte (NaCl) at several concentrations and verify the Onsager Equation. **(Conductometer)**
3.	determine the rate of the saponification of ethyl acetate at different temperatures conductometrically **(Conductometer)**
4.	measure the Mol. Wt. of Polymer using viscometer. **(Viscosity)**
5.	To verify the law of refraction for a given mixture of glycol + water system. **(Refractometer)**
6.	To determine the transition temperature of Glauber's salt by solubility method. **(Transition Temperature)**
7.	Determination of distribution coefficient of ammonia between Chloroform and water. **(Distribution Coefficient)**
8.	To determine the heat of the solution of the given acid by the solubility method. **(Heat of solution)**
9.	**Determination of the critical solution temperature (CST) of the phenol/water system & to study of the effect of additives on CST. (CST)**
10.	To determine the critical micelle concentration of anionic surfactant using a stalagmometer. **(CMC)**

C. Pedagogy

The students will be taught how to do experiments with instruments like pH meter, conductometer, polarimeter, potentiometer, Refractometer, and spectrophotometer for exercises in the physical chemistry laboratory. They will also perform non-instrumental experiments to understand the other concept of physical chemistry. Internal tests and viva will be taken as a part of continuous evaluation.

D. Learning Resources

1. Experimental Physical Chemistry by R. C. Das & B. Behera, (Tata McGraw hill Publishing Company Ltd., New Delhi)
2. Systematic Experimental Physical Chemistry by S. W. Rajbhoj and T. K. Chondhekar, (Anjali Publication, Aurangabad)
3. Advanced Practical Physical Chemistry by J. B. Yadav, (Goel Publishing House, Meerut)
4. Experimental Physical Chemistry by G. Peter Matthews, (Clarendon Press, Oxford, London)
5. Experimental Physical Chemistry by V. D. Athawale and Parul Mathur, (New Age International Publishers, New Delhi)
6. Advanced Physical Chemistry Experiments by Gurtu and Gurtu, (Pragati Prakashan, Meerut)

E. Course Outcomes (COs)

CO1	They will know the use of many simple instruments to extract valuable information and its Application
CO2	The student will learn about the experiments related to the properties solution and understand its concept.
CO3	Interpret data from various experiments, including the construction of appropriate graphs and evaluation of errors.
CO4	Develop skills in applying the distribution laws for various solutes between different appropriate solvent

F. Course Articulation Matrix

	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**
CO1											
CO2											
CO3											
CO4											

Correlation Levels: 1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

FACULTY OF MANAGEMENT STUDIES
DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES
HS106.02 (E):: ACADEMIC WRITING

I. Credits and Schemes:

Sem.	Course Code	Course Name	Credits	Teaching Scheme	Evaluation Scheme				
				Contact Hours/Week	Theory		Practical		Total
					Internal	External	Internal	External	
II	HS106.02 (E):	ACADEMIC WRITING	02	02	--	--	30	70	100

II. Course Outline

Module No.	Title / Topic	Classroom Contact Hours
1	**Academic Writing and Research Process** • *Introduction to Academic Writing* • *Academic Writing as a Part of Research* • *Types of Academic Writing* • *Features of Academic Writing* • *Importance of Good Academic Writing in various Academic Works*	**05**
2	**Anatomy of Academic Writing** • *Academic Vocabulary* • *Simple and Complex Sentences* • *Organizing Paragraphs* • *The Writing Process* • *Adopting Academic Writing Style*	**05**
3	**Key Academic Skills** • Note – taking • Note – making • Paraphrasing • Summarizing	**05**
4	**Accuracy in Academic Writing** • *Lexical Range* • *Academic Language and Structures*	**05**

	• *Elements of Writing* • *Proof Reading, Editing, and Rewriting*	
5	**Using and Citing Sources of Ideas** • *Academic Texts and their Types* • *Intellectual Honesty in Academic Writing* • *Avoiding Plagiarism – Idea Theft* • *Degrees of Plagiarism* • *Types of Borrowing* • *Anatomy of Citations* • *Common Citation Styles*	**05**
6	**Contemporary Practices in Academic Writing** • Analytical Essays • Graph / Table / Process Interpretation and Description • Writing Reports • Writing Research / Concept Papers	**05**
Total		**30**

III. Instruction Methods and Pedagogy

The course is based on practical learning. Teaching will be facilitated by reading material, discussion, task-based learning, projects, assignments and various interpersonal activities like writing, group work, independent and collaborative research, etc.

IV. **Evaluation**

The students will be evaluated continuously in the form of their consistent performance throughout the semester. There is no theoretical evaluation. There is just practical evaluation. The evaluation (practical) is schemed as 30 marks for internal evaluation and 70 marks for external evaluation.

Internal Evaluation

The students' performance in the course will be evaluated on a continuous basis through the following components:

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Paragraph Writing	1	3	03
2	Note-taking / Note-making	1	3	03
3	Paraphrasing / Summarizing	1	4	04
4	Essay Writing	1	5	05
5	Concept Paper Writing	1	10	10
5	Attendance and Class Participation			05
			Total	**30**

External Evaluation

The University Practical Examination will be for 70 marks and will test the professional communication skills and academic writing skills of the students.

Sl. No.	Component	Number	Marks per incidence	Total Marks
1	Viva / Practical /Quiz/ Project / Academic Writing	-	70	70
			Total	**70**

Course Outcome (COs):

After completion of the course, the student would:

CO1	have sound understanding of the concept and applications of academic writing
CO2	have acquired enough knowledge of academic writing style, strategy and approach
CO3	be able to demonstrate error free and effective academic writing
CO4	be able to demonstrate ability to work on project/report/paper writing
CO5	understand the concept of plagiarism and learn to use different citation styles as a part of referencing

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	-	-	-	-	-	3	-	-	-
CO2	-	-	-	-	-	-	-	3	-	-	-
CO3	-	-	-	-	-	-	-	3	-	-	-
CO4	-	-	-	-	-	-	-	3	-	-	-
CO5	-	-	-	-	-	-	3	-	-	-	-

Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

V. Reference Books / Reading

Essential Reading for Concepts

- Academic Writing for International Students, Routledge
- Academic Writing: A Guide for Management Students and Researchers. Monipally, M. M. & Pawar, B. S. Sage. 2010. New Delhi

Essential Reading for Activity and Teacher Resource

- ***Effective Academic Writing Level - 1,2,3,4 (Second Edition) By:*** Alice Savage, Patricia Mayer, Masoud Shafiei, Rhonda Liss, & Jason Davis; *Publisher: Oxford*

Additional Reading

- Writing Your Thesis (2nd Edition) by Paul Oliver, Sage
- Development Communication In Practice by Vilanilam V J, Sage
- Intercultural Communication by Mingsheng Li, Patel Fay, Sage
- www.owl.perdue.edu

OC 841: Applications of NMR and ESR Spectroscopy

Credit: 4 **Semester: 3**

A. Outline of the course:

Sr. No.	*Title of Unit*	*Minimum No. of hrs.*
1.	Introduction to Nuclear Magnetic Resonance (NMR) Spectroscopy	11
2.	Applications of NMR data	18
3.	Few important observations in NMR spectroscopy	11
4.	Basic theory of Electron Spin Resonance Spectroscopy and its applications	10
5.	Applications involving mainly IR,NMR and Mass spectroscopic data	10

B. Detailed syllabus:

Sr. No.	*Title of Unit*	*Minimum No. of hrs.*
1.	**Introduction to Nuclear Magnetic Resonance (NMR) Spectroscopy**	**11**
	Introduction, Magnetic properties of nuclei. Theory of NMR Spectroscopy. Chemical Shift. Factors contributing to the magnitude of Chemical Shift. Shielding Mechanism. Spin –Spin Splitting & its Mechanism. Vicinal coupling. Karpus equation. Coupling constant & conformation.	
2.	**Applications of NMR data**	**18**
	Applications of ^{1}H-NMR data (i). For elucidation of structure of organic molecules, (ii). Quantitative analysis, (iii).Study of hindered rotation & (iv) For inorganic chemistry problems: paramagnetic complexes, intramolecular tunneling process. Applications of ^{13}C NMR data (including DEPT) to elucidate molecular structure. 2D spectrum (COSY & HETCOR)	
3.	**Few important observations in NMR spectroscopy**	**11**
	Chemical exchange reaction reactions. Calculation of chemical shift in such cases. Simplification of complex spectra:	

	i. Shift reagents. ii. Spin decoupling iii. Deuterium labeling. Relaxation processes. Nuclear Overhauser Effect. Quantum mechanical treatment of spin- spins interaction between two nuclei.	
4.	**Basic theory of Electron Spin Resonance Spectroscopy and its applications**	**10**
	Introduction and basic theory of ESR. Presentation of the spectrum. Relaxation process and line width. Hyperfine splitting. Calculation of transition energies in some radicals. Analysis of Factors affecting the magnitude of g values. Zero field splitting and Kramers degeneracy. Role of ESR data in elucidation of molecular structure.	
5.	**Applications involving mainly IR,NMR and Mass spectroscopic data**	**10**
	Applications employing combined data from NMR,IR , Mass and UV spectroscopy for elucidation of molecular structure	

C. Pedagogy:

The topics will be discussed in interactive class room sessions using classical black-board teaching to power-point presentations. Interactive session in the form of seminars will be conducted by respective faculty members on weekly basis. Each student will give one seminar in a course. Problem solving sessions will also be conducted by respective faculty members on weekly basis. Course materials will be provided to the students from various primary and secondary sources of information. Internal test and quiz will be conducted as a part of continuous evaluation and suggestions will be given to students in order to improve their performance.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	The Programme aims at giving students certain important concepts to understand the fundamentals of NMR and ESR transitions
CO2	Ensuring that the students acquire good perceptions about NMR spectroscopy.
CO3	At the end, students would acquire sufficient knowledge to elucidate the structure of molecules from NMR, IR, Mass and UV spectroscopy.

E. Course Articulation matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	3	-	3	-	3	3	-	3

CO2	3	-	3	3	-	3	-	3	3	-	3
CO3	3	-	3	3	-	3	-	3	3	-	3

- Correlation Levels 1, 2 or 3 defined below
  1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

F. . References:

1. Fundamentals of Molecular Spectroscopy by C.N. Banwell & E.M. McCash, Publisher: Tata McGraw –Hill Publishing Company Ltd. New Delhi.
2. Physical Methods in Inorganic Chemistry by R.S .Drago Publisher : Affiliated East –West Pvt. Ltd.
3. Introduction to Magnetic Resonance Spectroscopy ESR,NMR, NQR by D. N. Sathyanarayana ; Publisher: IK Inernational Pvt Ltd.
4. Introduction to Spectroscopy by D.L. Pavia ,G.M. Lampman and G. S. Kriz, Publisher: Thomson Brooks.
5. Spectroscopy of organic compounds By P. S. Kalsi, Publisher: New Age International Pub.
6. Spectroscopic Identification of Organic Compounds by R.M. Silverstein & F. W. Webster and D. J .Kiemle. Publisher: John Wiley & Sons
7. Spectroscopic Methods in Organic Chemistry by D. H. Williams and I. Fleming, Publisher: McGraw –Hill Publishing Company
8. Elements of Magneto chemistry by R. L. Dutta & A. Syamal, Publisher: Affiliated East – West Press Pvt. Ltd.
9. Applications of absorption spectroscopy of organic compounds by J .C. Dyer, Publisher: Prentice – Hall of India Pvt.Ltd. New Delhi
10. An Introduction to the Analysis of Spin-Spin Splitting in High Resolution Nuclear Magnetic Resonance Spectra by J.D .Roberts. Publisher: W.A. Benjamin Inc. New York.

OC 842: Natural Product Chemistry

Credit: 4 Semester: 3

A. Outline of the course:

Sr. No.	*Title of Unit*	*Minimum No. of hrs.*
1.	Chemistry of Alkaloids and Steroids	12
2.	Chemistry of Flavonoids, Terpenoids and Carotenoids	12
3.	Chemistry of Purines and Pyrimidines	12
4.	Biosynthetic pathways of Natural Products	12
5.	Bioactive Marine Natural Products	12

B. Detailed syllabus:

Sr. No.	*Title of Unit*	*Minimum No. of hrs.*
1.	**Chemistry of Alkaloids and Steroids**	12
	Structure determination, stereochemistry and synthesis of some alkaloids and steroids. Chemistry of bile acids.	
2.	**Chemistry of Flavonoids, Terpenoids and Carotenoids**	12
	Structure determination, stereochemistry and synthesis of some flavonoids, terpenoids and carotenoids.	
3.	**Chemistry of Purines and Pyrimidines**	12
	Derivatives of purines and pyrimidines. Purines and pyrimidines in biochemical processes.	
4.	**Biosynthetic pathways of Natural Products**	12
	Acetate, Shikimate and Mevalonate pathways.	
5.	**Bioactive Marine Natural Products**	12
	Introduction, classification and importance of marine natural products.	

C. Pedagogy:
The topics will be discussed in interactive class room sessions using classical black-board teaching to power-point presentations. Interactive session in the form of seminars will be conducted by respective faculty members on weekly basis. Each student will give one seminar in a course. Course materials will be provided to the students from various primary and secondary sources of information. Internal test and quiz will be conducted as a part of continuous evaluation and suggestions will be given to students in order to improve their performance.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	Learners will gain the fundamental understanding of different classes of natural products and therapeutic importance.
CO2	Students will understand isolation, stereochemistry, and structure etermination of some natural products from alkaloids, steroids, flavonoids, terpenoids, and carotenoids
CO3	Students will acquire an understanding of biosynthetic pathways of natural products, the chemistry of purines and pyrimides and bioactive marine natural products

E. Course Articulation matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	-	-	3	-	-	3	-	3
CO2	3	-	3	-	-	3	-	-	3	-	3
CO3	3	-	3	-	-	3	-	-	-	-	3

- Correlation Levels 1, 2 or 3 defined below
  1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

F. References:

1. Organic Chemistry Vol 2 by I.L.Finar. Publisher: ELBS publication.
2. Chemistry of natural products by N. R .Krishaswamy, Publisher: University Press (India) Ltd. (1999).
3. The molecules of nature by J.B. Hendrickson. Publisher: W .A. Benjamin Inc (1965)
4. Natural products chemistry Vol I &II by K. Nakanishi et.al., Publisher: Academic press publications (1974).
5. Selected organic synthesis by I. Fleming. Publisher: John Wiley
6. Classics in organic synthesis by K.C. Nicolaou and E. J. Sorensen. Publisher : John Wiley

OC 843: Concepts of Heterocyclic Chemistry

Credit 4 **Semester 3**

A. Outline of the course:

Sr. No	Title of Unit	Minimum No. of Hrs
1	Nomenclature	08
2	Five membered heterocycles containing Nitrogen, Oxygen and Sulfur	12
3	Six membered heterocycles containing Nitrogen	10
4	Bicyclic hetrocycles containing Nitrogen	10
5	Heterocycles containing Oxygen	10
6	Chemistry of Indoles	10

B. Detailed Syllabus:

Sr. No	Title of Unit	Minimum No. of Hours
1.	**Nomenclature** Hantzsch- Widman Nomenclature For fused heterocycles.	08
2.	**Five membered heterocycles containing Nitrogen, Oxygen and Sulfur** Pyrrole: Reactions and Synthesis Furan: Reactions and Synthesis Thiophene: Reactions and Synthesis	12
3.	**Six membered heterocycles containing Nitrogen** Pyridines: Reactions and Synthesis Diazines-Pyridazine, Pyrimidine, and Pyrazine: Reactions and Synthesis	10
4.	**Bicyclic hetrocycles containing Nitrogen** Quinoline and Isoquinoline: Reactions and Synthesis	10
5.	**Heterocycles containing Oxygen** Pyryliums, 2- and 4-Pyrones : Reactions and Synthesis	10
6.	**Chemistry of Indoles** Indoles & Substituted Indoles: Reaction and Synthesis	10

C. Pedagogy:

The topics will be discussed in interactive class room sessions using classical black-board teaching to power –point presentations. Interactive session in the form of seminars will be conducted on weekly basis. Each student will give one seminar in a course. Problem solving sessions also will be conducted on weekly basis. Course materials will be provided to the students from various primary and secondary sources. Internal test and quiz will be conducted as a part of continuous evaluation and suggestions will be given to students to improve their performance whenever required.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	It is aimed at giving students sufficient information about heterocyclic compounds of biological importance
CO2	Ensuring that the students acquire knowledge to design the synthesis of heterocyclic compounds of choice

E. Course Articulation matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	3	-	3	-	-	3	-	3
CO2	3	-	3	3	-	3	-	-	3	-	3

- Correlation Levels 1, 2 or 3 defined below
  1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

F. Reference Books:

1. Heterocyclic chemistry by J. A. Joule and K. Mills , 5th edition Publisher: Blackwell Publishing Ltd. (2010).
2. Heterocyclic chemistry by Thomas L. Gilchrist 3rd edition Publisher: Prentice Hall (1997).
3. Handbook of Heterocyclic Chemistry by A. R. Kartrizky, C. A. Ramsden, J. A. Joule, ans V. Zhdankin 3rd edition Publisher: Elsevier (2010)
4. Name Reactions in Heterocyclic Chemistry by Jie Jack Li Publisher: Wiley Interscience

OC 844: Disconnection Approach for Organic Synthesis

Credit: 4 **Semester: 3**

A. Outline of the course:

Sr. No.	*Title of Unit*	*Minimum No. of hrs.*
1.	Introduction	09
2.	One and two group Disconnection	11
3.	Illogical two group disconnection	10
4.	Pericyclic reactions	10
5.	Ring synthesis and synthesis of acyclic and cyclic hetero compounds	10
6.	Miscellaneous topics	10

B. Detailed Syllabus:

Sr. No.	Title of Unit	Minimum No. of hrs
1.	**Introduction**	**09**
	Concept of synthon, synthetic equivalence, function group interconversion, concept and design of synthesis, criteria of good disconnection.	
2.	**One and two group Disconnection**	**11**
	One group disconnection: Disconnection and synthesis of alcohols, alkanes, olefins, simple ketones, acids and its derivative. **Two group disconnection:** Disconnection and synthesis of 1,3-dioxygenated skeletons, lactones, γ-halo ketones, β-hydroxyl carbonyl compounds, α, β-unsaturated carbonyl compounds, 1,3-dicarbonyls, 1,5- dicarbonyls and use of Mannich reaction.	
3.	**Illogical two group disconnection**	**10**
	Disconnection and synthesis of α and γ - hydroxyl carbonyl compounds, 1,2–diols, 1,4 and 1,6-dicarbonyl compounds.	
4.	**Pericyclic reactions**	**10**
	Disconnections based on Diels - alder reaction and its use in organic synthesis.	
5.	**Ring synthesis and synthesis of acyclic and cyclic hetero compounds**	**10**
	Special method for small ring preparations, synthesis of 3 and 4 member ring compounds. Synthesis of ethers, amines, nitrogen and oxygen containing acyclic and heterocyclic (five and six member) compounds.	
6.	**Miscellaneous topics**	**10**

	Protecting groups: Protection of organic functional groups, protecting reagents and removal of protecting groups. **Umpolung of reactivity**: Umpolung of carbonyl group, synthesis based on umpolung of carbonyl group–synthesis of 1,2 and 1,3-diketones, cyclic ketones etc. Synthesis of some complex organic molecules.	

C. Pedagogy:

The topics will be discussed in interactive class room sessions using classical black-board teaching to power-point presentations. Interactive session in the form of seminars will be conducted by respective faculty members on weekly basis. Each student will give one seminar in a course. Problem solving sessions will also be conducted by respective faculty members on weekly basis. Course materials will be provided to the students from various primary and secondary sources of information. Internal test and quiz will be conducted as a part of continuous evaluation and suggestions will be given to students in order to improve their performance.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	Ensuring that the students acquire better insight to design synthesis of molecules of choice.
CO2	At the end, students would acquire sufficient knowledge for future study/research.

E. Course Articulation matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	3	-	-	3	3	-	3	-
CO2	3	3	3	3	-	-	3	3	3	3	-

- Correlation Levels 1, 2 or 3 defined below
  1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

F. References:

1. Organic synthesis: the disconnection approach by Stuart Warren.
   Publisher: John Wiley and Sons (1994)
2. Selected Organic synthesis by Ian Fleming.
   Publisher: John Wiley and Sons (1977)
3. Organic synthesis: strategy and control by Paul Wyatt and Stuart Warren.
   Publisher: John Wiley and Sons (2007)
4. The logic of chemical synthesis by E. J. Corey and Xue-Min Chelg.
   Publisher: John Wiley and Sons.

OC 845: Practical

Credit: 7 **Semester: 3**

A. Outline of the Course:

Sr. No.	*Title of Unit*	*Minimum No. of hrs.*
1.	• Separation and identification of components in solid and liquid Organic mixtures. The mixture contains any three components from: ➢ Acid ➢ Phenol ➢ Base (Volatile/Non-volatile) ➢ Neutral (Volatile/Non-volatile)	**70**
2.	• Synthesis of dye intermediates, dyes and their application on fibers. Synthesis of dyes based on: ➢ Azo Dyes ➢ Azoic Dyes, ➢ Disperse Dyes, ➢ Acidic Dyes and etc. • Determination of molecular structure from spectral data. (^{1}H-NMR, IR, Mass and UV spectroscopic data). • Use of TG, DSC and HPLC.	**70**
3	• Visit to Industrial Organization/Institute that offers analytical services (component of curricular activity in practical course). The students will submit report of the visit to concerned teacher(s) in-charge for the laboratory work.	

B. Pedagogy:

Students will be made aware about various hazards in the laboratory and the safety measures for their prevention. Separation and identification of components in an organic mixture is very important for an organic chemist. Students will be first taught the chemistry of the separation and identification and they will demonstrate afterwards the separation and recognition of three components which will be from acid, phenol, base and neutral molecules. The science to synthesize dyes and its intermediates will be taught elaborately and if situation demands they will be shown with practical demonstration where they ought to be more careful. They will be told again how to utilize ^{1}H –NMR, IR, UV, Mass spectral data for structural determination. Students will gain hand-on experience to take measurements with instruments like thermal methods (TG, DSC), FTIR ,UV and HPLC and interpret the results. Students will be continuously evaluated through viva-voce for each experiment; each student will appear in internal test.

Visit to industrial organization/institute that provides analytical service is a component of the curricular for practical.

After the visit each student will submit a report under the guidance (if required) of the concerned teacher(s). The report will be evaluated.

C. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	Ensuring that students learn the science and art of organic synthesis which forms the back bone of semester III practical on more involved organic synthesis.
CO2	They will have the confidence of interpreting common spectral data of organic molecules

Course Articulation matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	-	3	-	3	-	-	3	3
CO2	-	3	3	3	-	-	-	-	-	-	3

- Correlation Levels 1, 2 or 3 defined below

  1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

E. References:

1. Comprehensive Practical Organic Chemistry By V K Alhuwalia and Sunita Dhingra, Publisher: University Press (India) Pvt Ltd.
2. Vogel's text book of quantitative analysis by J. Mendham, R. C. Denney, J.D .Barnes and M.J.K.Thomas. Publisher: Pearson Education Ltd.
3. Elementary practical of Organic Chemistry 5th edition, by A. I. Vogel Publisher: Longman group Ltd., London.
4. Spectrometric Identification of Organic Compounds, 6th edition by R. M. Silverstein,F.X Webster and D.J. Kiemle; Publisher: John Wiley & Sons.
5. Understanding Mass Spectra: A Basic Approach 2nd edition by R. Martin Smith. Publisher : John Wiley & Sons.

OC 846: Instrumental Methods of Analysis-I

Credit: 2 **Semester: 3**

A. Outline of the course:

Sr. No.	Title of Unit	Minimum No. of hrs.
1.	Nuclear Methods	06
2.	Electrophoresis	04
3.	TLC and HPTLC	05
4.	Ultra centrifugation	04
5.	Fluorescence and Phosphorescence	06

B. Detailed Syllabus:

Sr. No	Title of Unit	Minimum No. of hrs.
1.	**Nuclear Methods**	**06**
	Basic theory for applications of Activation analysis, Isotopic dilution method and Radiometric titrations to find the composition of a mixture.	
2.	**Electrophoresis**	**04**
	What is Electrophoresis? Understanding electrophoretic separation and applications of this technique. An account of Electro-osmotic flow.	
3.	**TLC and HPTLC**	**05**
	Basic principles involving TLC & HPTLC. Instruments and applications of these methods in the field of chemistry.	
4.	**Ultracentrifugation**	**04**
	Theory of centrifugation. Study of shape analysis of macromolecules and their conformational changes from experiments with ultracentrifuge. The importance of Ultracentrifugation for analysis of size distribution in polymers.	
5.	**Fluorescence and Phosphorescence Spectroscopy**	**06**
	Comparison of Phosphorescence and Fluorescence methods with emphasis on basic understanding of the theories. Factors affecting fluorescence and phosphorescence will be discussed. Few applications of these methods will be outlined .	

C. Pedagogy:

The topics will be discussed in interactive class room sessions using classical black-board teaching to power-point presentations. Interactive session in the form of seminars will be conducted by respective faculty members on weekly basis. Each student will give one seminar in a course. Problem solving sessions will also be conducted by respective faculty members on weekly basis. Course materials will be provided to the students from various primary and secondary sources of information. Internal test and quiz will be conducted as a part of continuous evaluation and suggestions will be given to students in order to improve their performance.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	Students acquire better perceptions about the importance of a variety of techniques to look methodically into different aspects of observable facts in chemistry.
CO2	It helps students to improve their understanding towards analytical methods and their applications in the field of chemistry

E. Course Articulation matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3		3						3	3
CO2	3		3		3						

- Correlation Levels 1, 2 or 3 defined below

  1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

E. References:

1. Essentials of nuclear chemistry by H. J. Arnikar. Publisher: New Age International, 1995.
2. Analytical Chemistry by Gary D. Christian Publisher: Wiley-India.
3. Fundamentals of Analytical Chemistry by D .A. Skoog , D. M. West, F. J. Holler and S. R. Crouch. Publisher: Brooks/Cole a part of Cengage Learning.
4. Electronic Absorption Spectroscopy and related Techniques by D. N. Sathyanarayana, Publisher: Universities Press, Hyderabad
5. Introduction to analytical ultracentrifugation by Greg Ralston Beckman Publisher: Beckman.
6. Analytical ultracentrifugation of polymers and nano particles by Walter Maechtle and Lars Borger, Publisher: Springer.
7. Analytical ultracentrifugation in biochemistry and polymer science by S.E. Hardings, A.J. Rowe and J.C. Horton eds. Royal Soc. Chem., 1992.

OC 847- Advance Problem solving in Chemical Sciences-I

Credit :2 **Semester -3**

A. Outline of the Course

Sr. No	Title of Unit	No of hr.
1	Organometallic Compounds	05
2	Electrochemistry & Solid State Chemistry	05
3	Chemical Kinetics & Thermodynamics	05
4	Stereochemistry	05
5	Organic Reaction Mechanism & Reactive Intermediates	05
6	IUPAC Nomenclature of Parent Hydride Compounds	05

B. Detailed Syllabus

Sr. NO.	Title of Unit and details	Number of hr.
1	Organometallic compounds	05
	Types of organometallic compounds .Classification of ligands and their uses. Applications of 18 electron rule. Importance of EAN, synthesis of organometallic compounds and its bonding, structure and reactivity The use of these compounds in homogeneous catalysis.	
2	Electrochemistry & Solid State Chemistry	05
	Different problems involving Electrochemistry & Solid State Chemistry will be taught.	
3	Chemical Kinetics & Thermodynamics	05
	Rate laws, Arrhenius equation, theory of reaction rates, determination of reaction mechanism. Fast reactions and experimental techniques for fast reaction. Numerical problems on thermodynamics.	
4	Stereochemistry	05
	Basic principles of stereochemistry, configurational and conformational isomerism in acyclic and cyclic compounds.	
5	Organic Reaction Mechanism & Reactive Intermediates	05
	Problems in Organic reaction mechanism involving addition, elimination and substitution reactions with electrophilic, nucleophilic or radical species. Certain methods for determination of reaction pathways.	

6	IUPAC Nomenclature of Parent Hydride Compounds Examples will be from Bridged hydrocarbons, Spiro compounds, Ring assemblies, Phane systems, Hetero hydrides compounds. Heteroatom replacement also will be considered for nomenclature.	05

C. Pedagogy:

The syllabus of this course has been framed keeping in mind the competitive examinations. The students will be exposed to problems based on topics included in the course. They will be trained to answer the questions asked in NET, GATE etc, and examinations. The topics will be discussed in interactive class room sessions using black board teaching to power-point presentations. Emphasis will be on solving problems. This course is hoped to equip the students to face the challenging questions in different examinations.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	The course aims at providing students the required skills to answer questions set in different competitive examinations
CO2	It also enhances the analytical skill and critical thinking of students towards theoretical concepts of chemistry

E. Course Articulation matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	-	-	-	-	3	-	-	-
CO2	3	-	3	-	-	-	-	3	-	-	-

- Correlation Levels 1, 2 or 3 defined below

  1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

F. References:

1. Principles of physical chemistry by B.R.Puri, L.R.Sharma and M.S. Pathania Publisher: Vishal Publishing Co., Jalandhar, Delhi.
2. Physical chemistry by P.W.Atkins and J.de.Paula Publisher: Oxford Press.
3. Chemical Applications of Group Theory by F.A.Cotton, Wiley Eastern Ltd. New Delhi, Bangalore.
4. Symmetry by R.McWeeny, Pergamon Press, Oxford, London, New York.
5. Group Theory and Symmetry in Chemistry by L.H.Hall, McGraw–Hill Book Company, New York, London.
6. Organic reactions and Mechanism by V. K. Ahluvalia and R. K. Parasher, Narosa Publishing House.
7. Organic reactions, Stereochemistry and Mechanism by P.S. Kalsi, New age Publishers.

8. Basic Organometallic Chemistry: Concepts, Synthesis and Applications by B.D. Gupta and A.J. Elias , CRC Press.
9. Systematic Nomenclature of Organic Compounds by S.Dhillon, Chinnappan Baskar, Shikha Baskar, Publisher: I.K. International Publishing House.
10. Systematic Nomenclature of Organic Chemistry: A Directory to Comprehension and Application of its Basic Principles by D.Hellwinkel, Publisher: Springer Science & Business Media.

OC 851: Medicinal Chemistry

Credit: 4 **Semester: 4**

A. Outline of the course:

Sr. No.	Title of Unit	Minimum No. of hrs.
1.	Drug design	09
2.	Pharmacokinetics	09
3.	Pharmacodynamics	06
4.	Cardiovascular drugs	12
5.	Antineoplastic agents	12
6.	Miscellaneous topics	12

B. Detailed syllabus:

Sr. No.	Title of Unit	Minimum No. of hrs.
1.	**Drug design**	**09**
	Introduction, Concept of lead, Analogues and Pro-drugs, An overview of drug design, development and discovery.	
2.	**Pharmacokinetic**s	**09**
	Drug absorption, drug distribution, drug metabolism (general pathway of drug metabolism, oxidative, reductive and hydrolytic reactions).	
3.	**Pharmacodynamics**	**06**
	receptors, chemical messengers, binding sites, receptor types and subtypes (protein receptors, DNA receptors with examples of agonists, partial agonists and antagonists	
4.	**Cardiovascular drugs**	**12**
	Anti-anginal and vasodilators, antihypertensive drugs, antiarrhythmic drugs, anti-cholesterolemic agents.	
5.	**Antineoplastic agents**	**12**
	Tumor cell properties, alkylating agents, antimetabolites, antibiotics, plant products, miscellaneous products.	
6.	**Miscellaneous topics**	**12**
	Antibiotics: General introduction, chemical classification, β-lactam antibiotics: penicillins, β-lactamase inhibitors, aminoglycosides, tetracyclines, chloramphenicol. **Psychoactive drug:** Antianxiety agents, antipsychotics, Antidepressants.	

C. Pedagogy:

The topics will be discussed in interactive class room sessions using classical black-board teaching to power-point presentations. Interactive session in the form of seminars will be conducted by respective faculty members on weekly basis. Each student will give one seminar in a course. Problem solving sessions will also be conducted by respective faculty members on weekly basis. Course materials will be provided to the students from various primary and secondary sources of information. Internal test and quiz will be conducted as a part of continuous evaluation and suggestions will be given to students in order to improve their performance.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	The Programme aims at giving students certain important concepts to understand the design of drugs
CO2	Ensuring that the students can recognize the role of drug absorption, drug-receptor interaction and metabolites to explain the mode of action of drugs
CO3	The students will gain knowledge about the chemistry of certain important drug molecules

E. Course Articulation matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	3	-	-	3	3	3	-	-
CO2	3	-	3	3	-	-	-	-	3	-	3
CO3	3	-	3	3	-	-	-	-	3	-	3

- Correlation Levels 1, 2 or 3 defined below

  1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

F. References:

1. Principles of Medicinal chemistry by William O-Foye, Thomas L.Lemke and David A. Williams, 4th edition, Publisher: B. I. Waverly Pvt., New Delhi (1995).
2. Essential of Medicinal Chemistry by Andrejus Korolkovas. Publisher: Wiley –India edition (1988).
3. Instant Notes: Medicinal Chemistry, edited by Graham L . Patric, Publisher: Viva Books Pvt. LTd (2002)
4. Medicinal Chemistry by Ashutosh Kar, Publisher: New Age international (P) Ltd. (2005)
5. Wilson and Griswold's Textbook of Organic Medicinal and pharmaceutical Chemistry by John H. Block and John M. Beale, Publisher: Lippincott Williams &Wilkins (2004)
6. Textbook of Medicinal Chemistry Vol I & II by V. Alagarsamy, Publisher: Elsevier (2010).
7. An Introduction of Medicinal Chemistry by Graham L.Patrick Publisher: Oxford University Press (2009).

OC 852: Chemistry of Synthetic Dyes

Credit: 4 **Semester: 4**

A. Outline of the course:

Sr. No.	*Title of Unit*	*Minimum No. of hrs.*
1.	Introduction	12
2.	Brief account of Near Infrared Absorption (NIR) dyes	08
3.	functional dyes	08
4.	Specific Azo dyes	12
5.	Non textile dyes	12
6.	Some aspects of fluorescent brightening agents	08

B. Detailed Syllabus:

Sr. No	*Title of Units*	*Minimum No .of hrs.*
1.	**Introduction**	**12**
	Importance of Chemical chromophors in dyes. Classification of dyes. Brief description of each class and their principle applications.	
2.	**Brief account of Near Infrared Absorption (NIR) dyes**	**08**
	Introduction of NIR dyes. Importance of Cyanine type chromophores, donor-acceptor chromophores. Applications of NIR dye in many areas.	
3.	**Functional dyes**	**08**
	Important features and synthesis of different functional dyes such as indigoid dyes, reactive dyes, anthraquinone dyes etc. and their various applications.	
4.	**Specific Azo dyes**	**12**
	Learning specific azo dyes which are colorants for high technology applications. The relation between color and constitution of Azo dyes will be elaborately discussed.	
5.	**Non textile dyes**	**12**
	Characteristics of non-textile dyes, their importance and applications.	
6.	**Some aspects of fluorescent brightening agents**	**08**
	Characteristics of brightening agents, its status and applications.	

C. Pedagogy:

Subject matters will be discussed in interactive class room sessions using classical black-board teaching to power-point presentations. Interactive session in the form of seminars will be conducted by respective faculty members on weekly basis. Each student will give one seminar in a course. Course materials will be provided to the students from various primary and secondary sources of information. Internal test and quiz will be conducted as a part of continuous evaluation and suggestions will be given to students in order to improve their performance.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	It is aimed to provide students important concepts to design and synthesize dyes, pigments and brightening agents for specific use.
CO2	Ensuring that the students can identify the importance of non-textile dyes
CO3	The students will gain knowledge about the basic theory of imparting color by dyes.

E. Course Articulation matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	3	-	-	-	3	-	-	-
CO2	3	-	3	3	-	-	-	-	-	-	-
CO3	3	-	3	3	-	-	3	-	-	-	-

- Correlation Levels 1, 2 or 3 defined below

  1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

F. References:

1. Colour Chemistry: Synthesis, properties and applications of organic dyes and pigments by Heinrich Zollinger, Publisher: VCH, Germany (1987).
2. Advances in colour chemistry series, vol 3 (Modern colorants, synthesis and structure) edited by A.T. Peters and H. S. Freeman, Publisher: Blackie Academic and Professional.
3. The chemistry of synthetic dyes and pigments by H. A. Lubs, Publisher: Reinhold Publication (1955).
4. The production and applications of fluorescence brightening agents by Milos Zahradnik, Publisher: John Wiley & Sons 1982.
5. Infrared absorbing dyes edited by Masaru Matsuoka, Publisher: Plenum Press 1990.

OC 853: Energy & Wave Function of Molecular Orbital in Organic Chemistry

Credit: 4 **Semester: 4**

A. Outline of the course:

Sr. No.	Title of Unit	Minimum No. of hrs.
1.	Introduction to Huckel Molecular Orbital Method	10
2.	Molecular Symmetry for Simplification of Secular Determinants	10
3.	Extensions, modifications and applications of Huckel approach	12
4.	First Order Perturbation	10
5.	Advanced MO methods	10
6.	Linear Free Energy	08

B. Detailed Syllabus:

Sr. No.	Title of unit	Minimum No. of hrs.
1.	**Introduction to Huckel Molecular Orbital Method**	**10**
	Introduction Basis of the method. iii. Huckel determinant for simple carbon π systems, determination of molecular orbital energies & LCAO-MO coefficients. Calculation of Delocalization energy and its importance. iv. Symmetry properties of Huckel molecular orbitals.	
2.	**Molecular Symmetry for Simplification of Secular Determinants**	**10**
	Transforming secular determinants which are derived from atomic orbital into secular determinants which are expressed in terms of group (symmetry) orbitals. Formal use of symmetry by means of group theory to obtain energy of MOs. iii. Simple method to obtain molecular orbital energies for un- branched cyclic and acyclic π systems	
3.	**Extensions, Modifications and Applications of Huckel Approach**	**12**
	i. Calculation for molecules containing heteroatoms. Inclusion of overlap	

	ii. Properties of alternant hydrocarbons, Dewar nonbonding MO (NBMO) method. Uses of NBMO coefficients. iii. Huckel Theory for cyclic conjugated molecules & aromaticity.	
4.	**First Order Perturbation**	**10**
	i. Introduction to first order perturbation calculation. Relationship between total π electronic energy, bond order etc. . Systematic approach to first order perturbation calculation, generalized bond orders. First order perturbation calculation with degenerate eigenvalues. Perturbation parameters for heteroatoms.	
5.	**Advanced MO Methods**	**10**
	i. Advanced MO methods: Polyelectron wave function, Slater determinants, Hartree Fock equation, ZDO approximation, Pariser Parr Pople calculation for simple π systems. Simple all valence calculation using ZDO approximation.	
6.	**Linear Free Energy**	**08**
	Hammett linear free energy relationship and its various applications.	

C. Pedagogy:

Subject matters will be discussed in interactive class room sessions using classical black-board teaching to power-point presentations. Interactive session in the form of seminars will be conducted by respective faculty members on weekly basis. Each student will give one seminar in a course. Problem solving sessions will also be conducted by respective faculty members on weekly basis. Course materials will be provided to the students from various primary and secondary sources of information. Internal test and quiz will be conducted as a part of continuous evaluation and suggestions will be given to students in order to improve their performance.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	It is aimed that the students will recognize the remarkable predictive power of two simple methods, Huckel & first order perturbation.
CO2	Ensuring that the students can appreciate the assumptions involved for improved molecular orbital calculations like Pariser Parr Pople calculations etc.
CO3	The students will identify the importance of the power of Hammett linear free

	energy relationship to predict behavior different chemical reactions

- **Course Articulation matrix**

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	3	-	-	3	3	-	-	3
CO2	3	3	3	3	-	-	3	3	-	-	3
CO3	3	3	3	3	-	-	3	3	-	-	3

- Correlation Levels 1, 2 or 3 defined below

  1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

` **E. References:**

1. Molecular Orbital theory of organic Chemistry by M. J. S. Dewar, Publisher: McGraw – Hill, New York.
2. Molecular Orbital theory for organic Chemists by A.Streitwieser, Publisher: Wiley, New York
3. Valence by C.A. Coulson, Publisher: Oxford University Press, London and New York.
4. Chemical applications of group theory by F.A.Cotton, Publisher: Wiley Interscience, New York.
5. Quantum mechanics for Organic Chemists by Howard E. Zimmerman, Publisher: Academic Press, New York.
6. Approximate Molecular orbital Theory by J.Pople and D.L.Beveridge, Publisher: McGraw Hill , New York.
7. Group Theory and Symmetry in Chemistry by L.H.Hall Publisher: McGraw –Hill Book Company.

OC 854: Selected Topics in Organic Chemistry

Credit: 4 **Semester: 4**

A. Outline of the course:

Sr. No.	Title of Unit	Minimum No. of hrs.
1.	Concepts of Stereochemistry	10
2.	Asymmetric Synthesis	10
3.	Pericyclic Reactions	10
4.	Catalysis	10
5.	Physical Methods for Organic Reactions	10
6.	Miscellaneous	10

B. Detailed Syllabus:

Sr. No	Title of Unit	Minimum No. of hrs.
1.	**Concepts of Stereochemistry**	**10**
	Conformational Analysis of acyclic, cyclic and Fused Systems	
2.	**Asymmetric Synthesis**	**10**
	Resolution – Optical and Kinetic Determination of enantiomeric /diastereomeric excess Methods of asymmetric induction-substrate, reagents and catalyst controlled reactions	
3.	**Pericyclic Reactions**	**10**
	Electrocyclic, Cycloaddition, and Sigmatropic Reactions Principles and Application of Photochemical reaction in Organic Chemistry	
4.	**Catalysis**	**10**
	Introduction, The Basics of catalysis, Homogeneous catalysis, Heterogeneous catalysis and Biocatalysis	
5.	**Physical Methods for Organic Reactions**	**09**
	Microwave assisted organic synthesis: Theory and Applications Ultrasound assisted organic synthesis: Theory and Applications	
6.	**Miscellaneous**	**11**
	Introduction to Environmental Chemistry Synthesis of Vitamins (Vitamin B_1, B_6, C and Biotin)	

C. Pedagogy:

Subject matters will be discussed in interactive class room sessions using classical black-board teaching to power-point presentations. Interactive session in the form of seminars will be conducted by respective faculty members on weekly basis. Each student will give one seminar in a course. Course materials will be provided to the students from various primary and secondary sources of information. Internal test and quiz will be conducted as a part of continuous evaluation and suggestions will be given to students in order to improve their performance.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	The topics are so designed to provide students important concepts on conformational analysis and how do conformations govern chemical reactions/properties.
CO2	Ensuring that the students can recognize the importance of non-conventional synthesis using microwave and ultrasound
CO3	The students will gain knowledge about the importance of Asymmetric synthesis
CO4	The dangers posed by water and air pollution and remedial measures will be discussed and this will help the students to address the problems which common citizen face

E. Course Articulation matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	-	-	-	-	-	-	-	-
CO2	3	-	3	3	-	3	-	-	3	-	3
CO3	3	3	3	3	-	3	-	-	3	-	3
CO4	3	-	-	-	-	-	3	-	-	3	-

- Correlation Levels 1, 2 or 3 defined below

  1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

F. References:

1. Practical Microwave synthesis for Organic Chemist by C. Oliver Kappe, Publisher: Wiely-VCH.
2. Review “Ultrasound in Synthetic Organic Chemistry” by Timothy J. Mason. Chemical Society Reviews1997, Vol. 26, Page No. 443-451.
3. Stereochemistry of carbon compounds by E. L. Eliel, Publisher: McGraw Hill.
4. Stereochemistry and mechanism through solved problems by P .S. Kalsi, Publisher: Wiley Eastern Ltd.
5. Stereochemistry: Conformation and Mechanism P .S. Kalsi, Publisher: New Age International.
6. Stereochemistry of organic compounds by D. Nasipuri, Publisher: New Academic Sciences.
7. Organic Chemistry Vol 2 by I. L. Finar, Publisher: ELBS Publication.
8. Environmental Chemistry by Samir K Banerji, Publisher: Prentice –Hall of India Pvt. Ltd
9. Catalysis: Concepts and Green Applications by Gadi Rothenberg, Publisher: Weily-VCH Verlag GmbH & Co.

OC 855: Practical

Credit: 7 **Semester: 4**

A. Outline of the Course:

Sr. No.	**Title of Unit**	**Minimum No. of hrs.**
1.	• Multiple step synthesis of Heterocyclic compounds involving: Quinazoline, oxazole, 69uinolone, Benzimidazole, Benzotriazole, Thiazole, Benzothiazole, Indole, Chromone and Coumarin etc. • Synthesis of some drugs intermediates likes: Aspirin, Salol, Yara-yara, Ethyl-cinnamate, Hydantion, Substituted sulphonamide, Butesin, Acridone and etc.	**70**
2.	• Structure determination from spectral data (Spectra were characterized on the basis of ^{1}H-NMR ^{13}C-NMR, DEPT, APT, IR, Mass and UV spectroscopy). • Experiment on Paper chromatography. (Separations of dyes in acidic and basic medium) • One experiment on monitoring the synthesis of organic compounds by utilizing thin layer and column chromatography.	**35**
3.	• % Estimation of Nitrogen by Kjeldahl's method. • Determination of saponification value of oils. • Determination of Reichert-Meisel value. • Experiments base on Protection and deprotection by using BOC and TMS. • Determination of amount of amino compounds using starch-iodide paper as a external indicator.	**35**
4.	• Visit to Industrial Organization/Institute that offers analytical services (component of curricular activity in practical course). • Students will submit report of the visit to concerned teacher(s) in-charge for the laboratory work.	

B. Pedagogy:

The students will be reminded about various hazards in the laboratory and the safety measures for their prevention.

They will be explained at the beginning what care to be taken while performing more involved synthesis of heterocyclic compounds and drug intermediates. If necessary they will get necessary practical demonstration of the required synthesis. In continuation of semester III practical involving spectral data, students will be analyzing more difficult spectral data in this semester. Use of different chromatographic techniques like paper, thin layer and column will be explained with demonstration and then students will need to perform the experiments. Students will be given suitable experiments to demonstrate the advantages and disadvantages of Kjeldahl's method for determination of nitrogen. Students will be continuously evaluated through viva-voce for each experiment and one internal test at the completion of the practical course.

Visit to industrial organization/institute that provides analytical service is an element of the curricular for practical. It is mandatory for each student to submit a report of the visit under the guidance (if required) of the concerned teacher(s). The report will be evaluated.

C. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	To develop sound knowledge of organic synthesis through exhaustive practical training
CO2	To be able to characterize the molecules from spectral data
CO3	Students will acquire sufficient knowledge and confidence to work in industry, carry out further study/research or join academic institutions

D. Course Articulation matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	3	3	-	3	-	3	-	-	3	3
CO2	-	3	3	3	-	-	-	-	-	-	3
CO3	3	3		3	3		3			3	3

- Correlation Levels 1, 2 or 3 defined below
  1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

E. References:

1. Comprehensive Practical Organic Chemistry By V K Alhuwalia and Sunita Dhingra, Publisher: University Press (India) Pvt Ltd.
2. Vogel's text book of quantitative analysis by J.Mendham, R .C .Denney, J. D. Barnes and M.J.K.Thomas; Publisher: Pearson Education Ltd.
3. Elementary practical of Organic Chemistry 5th edition, by A.I.Vogel Publisher: Longman group Ltd., London.
4. Spectrometric Identification of Organic Compounds, 6th edition by R. M. Silverstein, F.X Webster and D.J. Kiemle . Publisher: John Wiley & Sons.
5. Understanding Mass Spectra: A Basic Approach, 2nd edition by R. Martin Smith. Publisher: John Wiley & Sons.
6. Organic synthesis: strategy and control by Paul Wyatt and Stuart Warren, Publisher: John Wiley and Sons (2007).

OC 856: Instrumental Methods of Analysis-II

Credits: 2 **Semester: 4**

A. Outline of the course.

Sr. No.	Title of Unit	Minimum No. of hrs.
1.	Electroanalytical Chemistry	04
2.	Potentiometry	03
3.	Polarography	03
4.	Applications of Spectroscopic Data	10
5	Solubility equilibria & Precipitation Titrations	05

B. **Detailed syllabus:**

Sr. No.	Title of Unit	Minimum No of hrs.
1.	**Electroanalytical Chemistry**	04
	Electrochemical Cells, Galvanic and Electrolytic Cells, Cells without liquid junctions. Electrical double layer, Faradaic and non-Faradaic currents. Thermodynamics of cell potential, Liquid junction potential, polarization. Some numerical problems.	
2.	**Potentiometry**	04
	General Principles, Reference electrodes, Metallic indicator electrodes--- electrodes of first, second and third kind ,metallic redox indicators .Membrane indicator electrode .Biosensors and its applications in clinical analysis. Potentiometric titrations.	
3.	**Polarography**	03
	Important features, diffusion current. Ilkovic equation. Half-wave potential. Applications.	
4.	**Applications of Spectroscopic Data**	10
	The structure of molecule will be elucidated from IR, NMR and mass spectroscopic data.	
5.	**Solubility equilibria &Precipitation titrations**	04
	Feasibility of precipitation titrations, indicators for precipitation titrations, Fajans method.	

C. Pedagogy:

Subject matters will be discussed in interactive class room sessions using classical black-board teaching to power-point presentations whenever required. Interactive session in the form of seminars will be conducted by respective faculty members on weekly basis. Each student will give one seminar in a course. Course materials will be provided to the students from various primary and secondary sources of information. Internal test and quiz will be conducted as a part of continuous evaluation and suggestions will be given to students in order to improve their performance.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	The topics are so designed to provide students important basic concepts on statistical analysis and how to use the tools to analyze data
CO2	Ensuring that the students can employ spectroscopic data rationally to predict the molecular structure
CO3	The significance of solubility equilibria, precipitation titrations & Polarographic data to understand many chemical situations will be realized

E. Course Articulation matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3			3			3				3
CO2	3	3	3								
CO3	3										3

- Correlation Levels 1, 2 or 3 defined below

  1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

F. References:

1. Quantitative Analysis by R .A .Day and A. L. Underwood Publisher: Prentice-Hall of India, Pvt. Ltd. New-Delhi-110001.

2 . Analytical Chemistry by Gary D. Christian Publisher: Wiley-India.

3. Instrumental Methods of Chemical Analysis by Gurdeep R .Chatwal and Sham K. Anand Publisher: Himalaya Publishing House, Mumbai.

4. Analytical Chemistry Principals by John H.Kennedy Publisher: Saunders College Publishing, New York.

5. Introduction to Spectroscopy by D.L. Pavia , G .M. Lampman and G.S.Kriz, Publisher: Thomson Brooks.

6. Spectroscopy of organic compounds By P. S. Kalsi, Publisher: New Age International Publishers.

7. Spectroscopic Identification of Organic Compounds by R.M. Silverstein, F.X. Webster and D.J. Kiemle . Publisher: John Wiley & Sons.

8. Spectroscopic Methods in Organic Chemistry by D.H. Williams and I. Fleming, Publisher: McGraw – Hill Publishing Company.

9. Applications of absorption spectroscopy of organic compounds by J.C.Dyer, Publisher: Prentice –Hall of India Pvt. Ltd. New Delhi.

10. Instrumental Analysis by D. A. Skoog , F. J. Holler and S.R. Crouch, Publisher: Cengage Learning (Indian Edition).

OC 857- Advance Problem solving in Chemical Sciences-II

Credit :2 **Semester -4**

A. Outline of the Course

Sr. No.	Title of Unit	No of hr.
1	Problems Based on Spectroscopic Data	06
2	Disconnection Approach	06
3	Specific Organic Name Reaction & Reagents	08
4	Heterocyclic Chemistry	06
5	Chemistry of s & p block elements	04

B. Detailed Syllabus

Sr. NO	Title of Unit and details	Number of hr.
1	**Problems Based on Spectroscopic Data**	06
	Molecular structure determination from UV-visible, Mass, IR &, NMR data.	
2	**Disconnection Approach**	06
	One & two group disconnection, illogical two group disconnection will be deliberated with examples commonly asked.	
3	**Specific Organic Name Reaction & Reagents**	08
	Chemistry of common Name reactions and rearrangements will be discoursed. Problems in applications of these in organic synthesis will be considered.	
4	**Heterocyclic Chemistry**	06
	Problems concerning synthesis and reactivity of common heterocyclic compounds containing one or two heteroatoms (O,N,S) will be discussed.	
5	**Chemistry of s & p block elements**	04
	Problems based on s & p block elements and their compounds will be considered.	

C. Pedagogy:

The syllabus of this course has been framed keeping in mind the competitive examinations. The students will be exposed to problems based on topics included in the course. They will be trained to answer the questions asked in NET, GATE ete, examinations. The topics will be discussed in interactive class room sessions using black board teaching to power-point presentations. Emphasis will be on solving problems. This course is hoped to equip the students to face the challenging questions in different examinations.

D. Course Outcomes (COs):

At the end of the course, the students will be able to

CO1	The course aims at providing students the required skills to answer questions set in different competitive examinations
CO2	It also enhances the analytical skill and critical thinking of students towards theoretical concepts of chemistry.

E. Course Articulation matrix

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	3	-	-	-	-	3	-	-	-
CO2	3	-	3	-	-	-	-	3	-	-	-

- Correlation Levels 1, 2 or 3 defined below

  1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

F. References:

1. Spectroscopy of organic compounds by P.S.Kalsi Publisher: New Age International Publishers.
2. Physical Methods in Inorganic Chemistry by R.S .Drago Publisher : Affiliated East –West Pvt. Ltd.
3. Introduction to Magnetic Resonance Spectroscopy ESR, NMR, NQR by D. N. Sathyanarayana ; Publisher: IK Inernational Pvt Ltd.
4. Introduction to Spectroscopy by D.L. Pavia ,G.M. Lampman and G. S. Kriz, Publisher: Thomson Brooks.
5. Spectroscopic Identification of Organic Compounds by R.M. Silverstein & F. W. Webster and D. J .Kiemle. Publisher: John Wiley & Sons.
6. Heterocyclic chemistry by J. A. Joule and K. Mills , 5th edition, Publisher: Blackwell Publishing Ltd. (2010).
7. Heterocyclic chemistry by Thomas L. Gilchrist 3rd edition, Publisher: Prentice Hall (1997).
8. Organic synthesis: the disconnection approach by Stuart Warren. Publisher: John Wiley and Sons (1994).
9. The logic of chemical synthesis by E. J. Corey and Xue-Min Chelg. Publisher: John Wiley and Sons.
10. Name reactions Volume I and II by Ji Jack Li, Wiley publishing
11. Name Reactions and Reagents by B. P. Mundy, Michael G. Ellerd and F. G. Favaloro, Wiley-Interscience publisher.
12. Periodicity and the s & p block elements by N.C. Norman , Oxford Press.